Name
Date
Course

Reflection, Symmetry, and Inversion

Relation: A connection between two values (usually called X and Y).

Three Ways to Represent a Relation

Examples

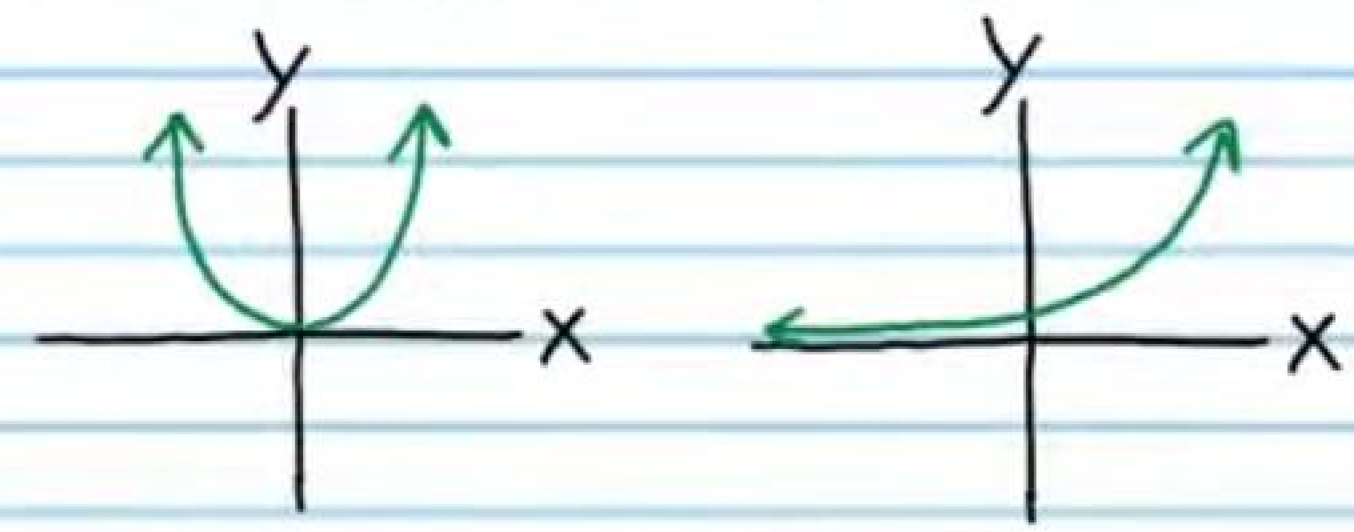
I. Equation — $y=x^2$ — $y=3^x$

II. Table

X	-2	-1	0	1	2
Y	4	1	0	1	4

X	-2	-1	0	1	2
Y	1/9	1/3	1	3	9

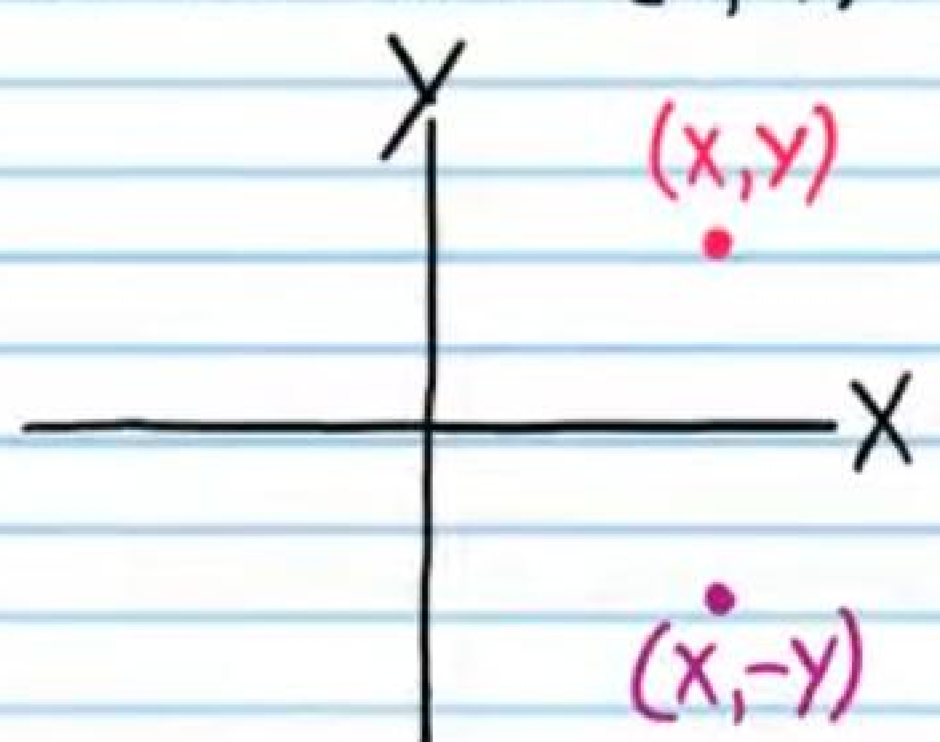
III. Graph



Reflection

Reflection across the X-Axis

If (x,y) is a point on the xy-plane, then the reflection of this point across the x-axis is the point with coordinates $(x,-y)$.



Relations are written below in table form. Reflect each point across the x-axis, and write the coordinates in the tables at the right.

① Reflect across the x-axis

X	2	5	-1	0	4
Y	1	-3	7	-2	4

X	2	5	-1	0	4
Y	-1	3	-7	2	-4

② Reflect across the x-axis

X	0	4	17	-3	-5
Y	-1	8	20	-8	-10

X	0	4	17	-3	-5
Y	1	-8	-20	8	10

③ Reflect across the x-axis

X	9	4	6	3	0
Y	-9	-4	10	7	2

X	9	4	6	3	0
Y	9	4	-10	-7	-2

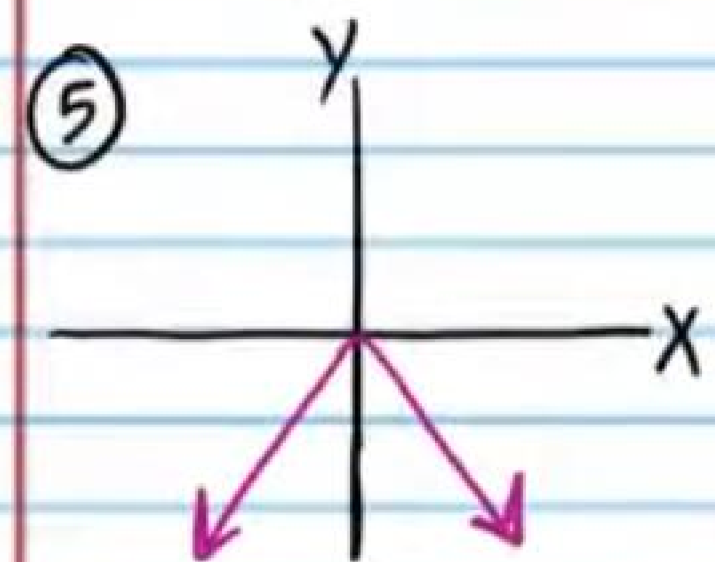
④ Reflect across the x-axis

X	-6	4	3	3/4	10
Y	0.1	1/2	6	0	-8

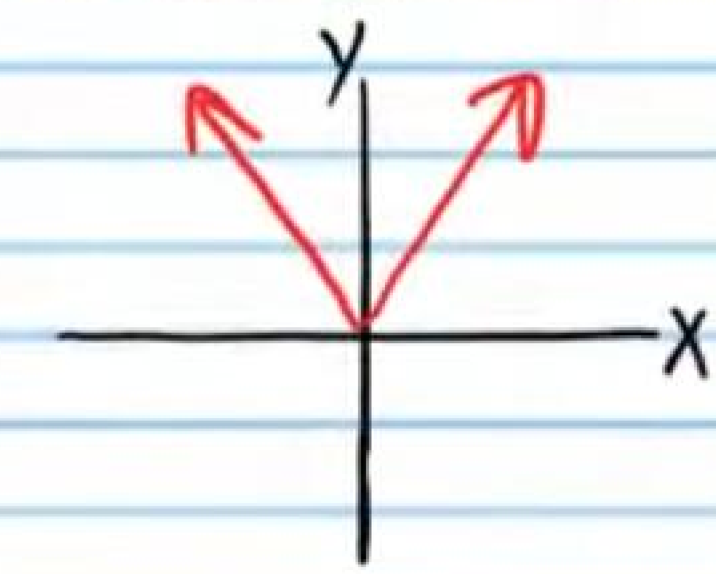
X	-6	4	3	3/4	10
Y	-0.1	$-\frac{1}{2}$	-6	0	8

Relations are written below in the form of graphs. Draw the reflection of each graph across the x-axis.

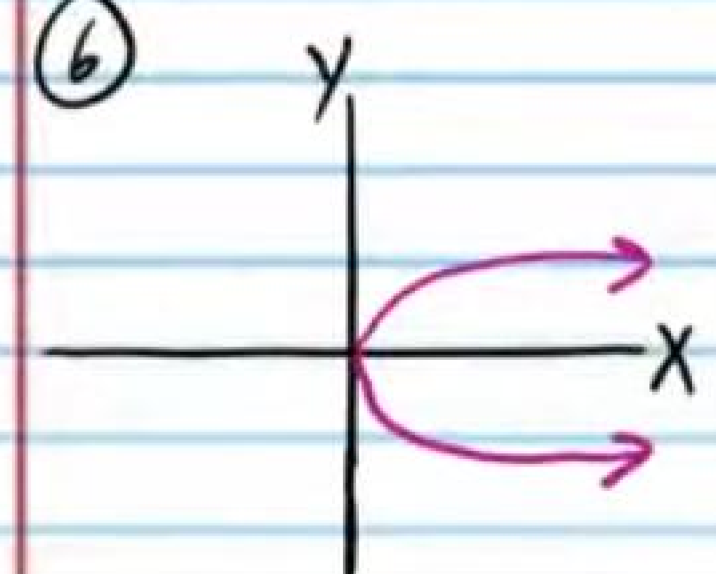
⑤



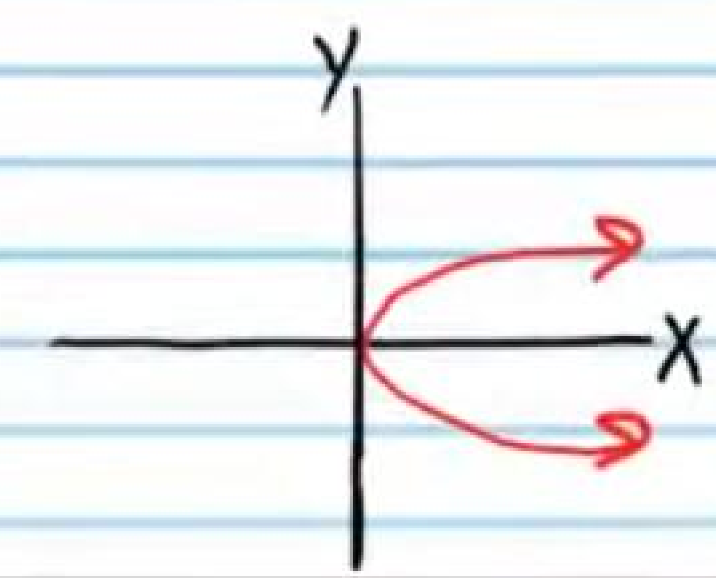
Reflect across the x-axis



⑥



Reflect across the x-axis



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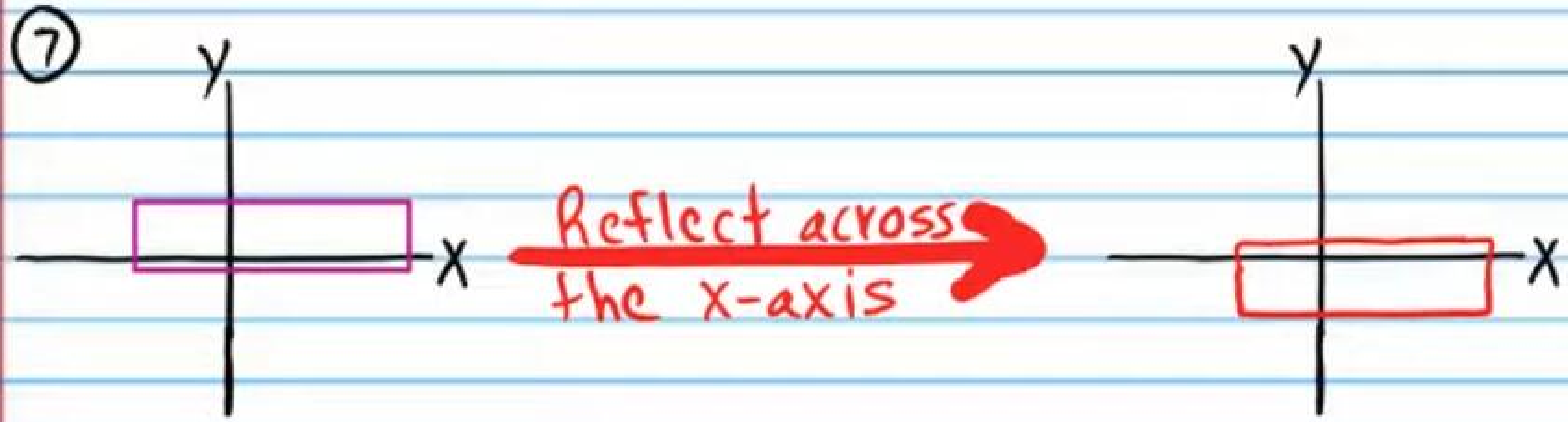

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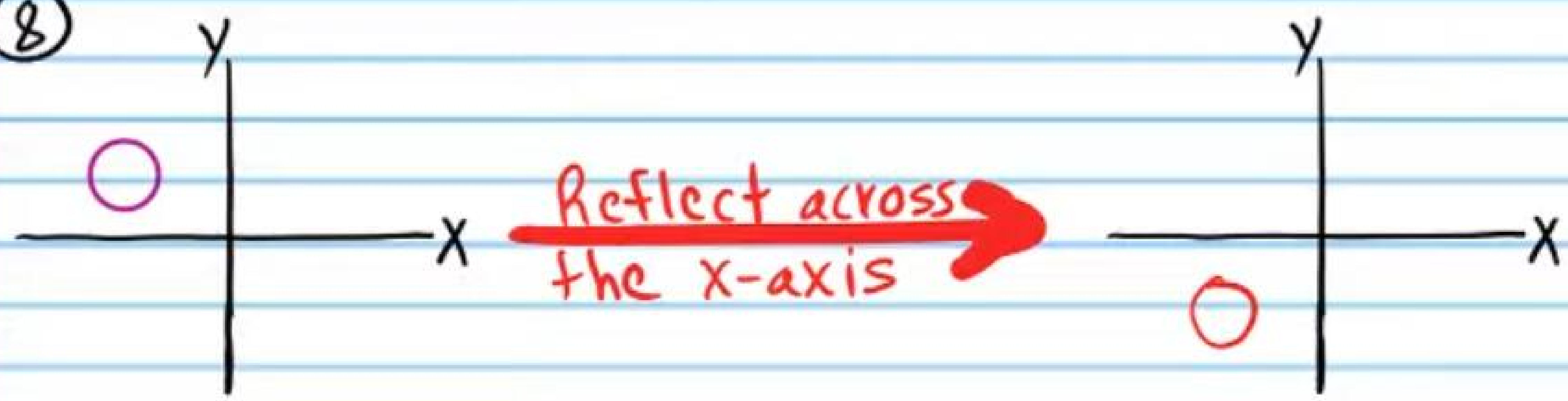

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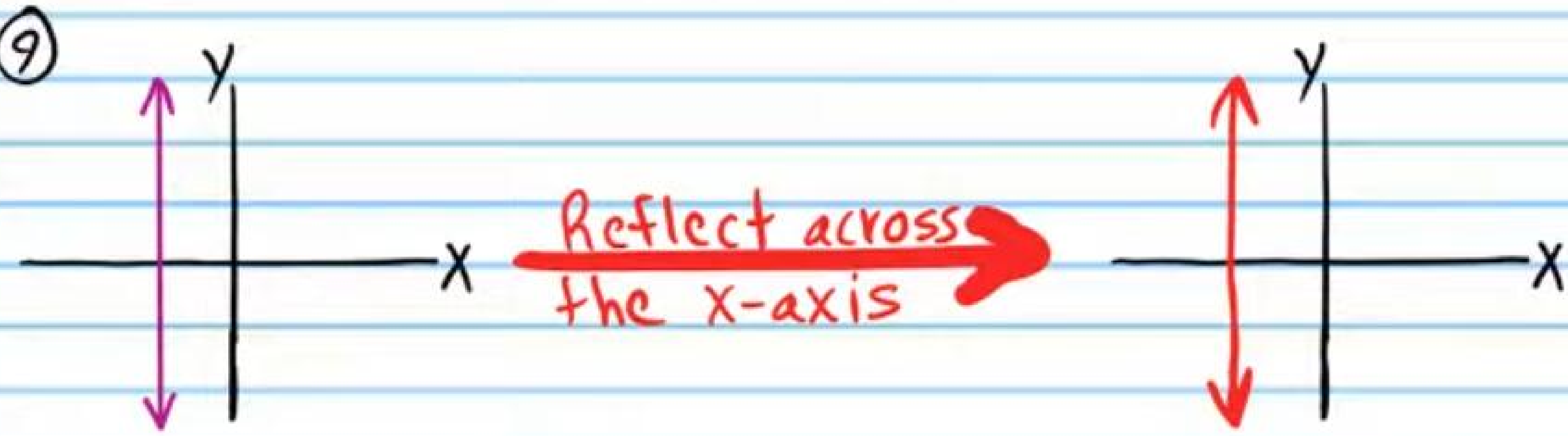

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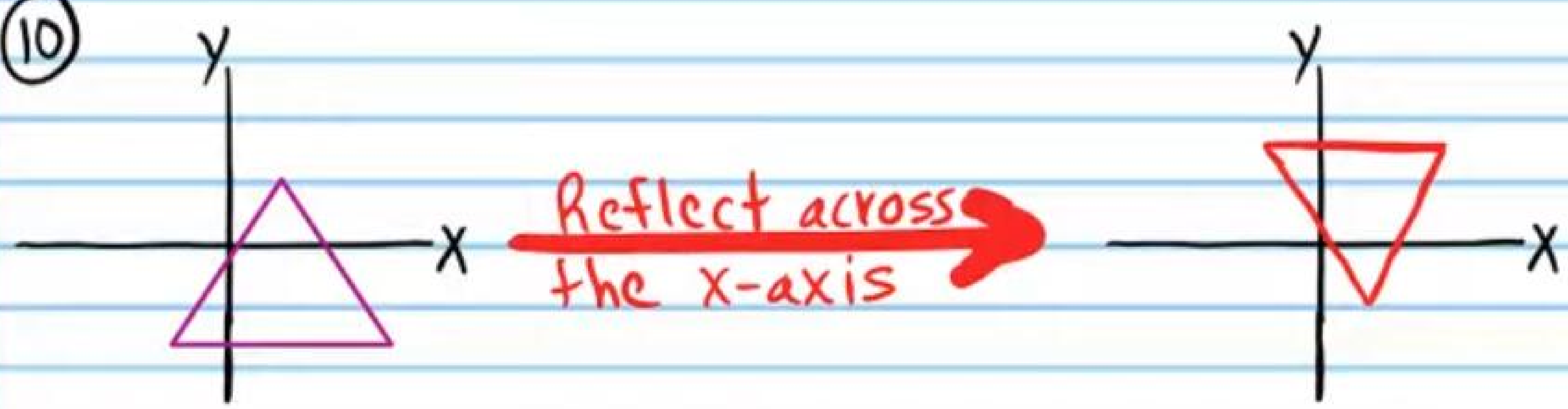


The equations of relations are written below. Write equations of reflections across the x-axis. Simplify if possible.

(11) $x = y - 2 \xrightarrow[\text{the x-axis}]{\text{Reflect across}} \boxed{x = -y - 2}$

(12) $y^3 = x^2 + 1 \xrightarrow[\text{the x-axis}]{\text{Reflect across}} (-y)^3 = x^2 + 1$

$\boxed{-y^3 = x^2 + 1}$

(13) $y^2 = \sin(x+10) \xrightarrow[\text{the x-axis}]{\text{Reflect across}} (-y)^2 = \sin(x+10)$

$\boxed{y^2 = \sin(x+10)}$

(14) $\log_2 x = \sqrt{y} \xrightarrow[\text{the x-axis}]{\text{Reflect across}} \boxed{\log_2 x = \sqrt{-y}}$

(15) $y = x^5 \xrightarrow[\text{the x-axis}]{\text{Reflect across}} \boxed{-y = x^5}$

(16) $3^{y-2} = x - 6 \xrightarrow[\text{the x-axis}]{\text{Reflect across}} \boxed{3^{-y-2} = x - 6}$

(17) $\cot y^4 = 1 + |x| \xrightarrow[\text{the x-axis}]{\text{Reflect across}} \cot(-y)^4 = 1 + |x|$

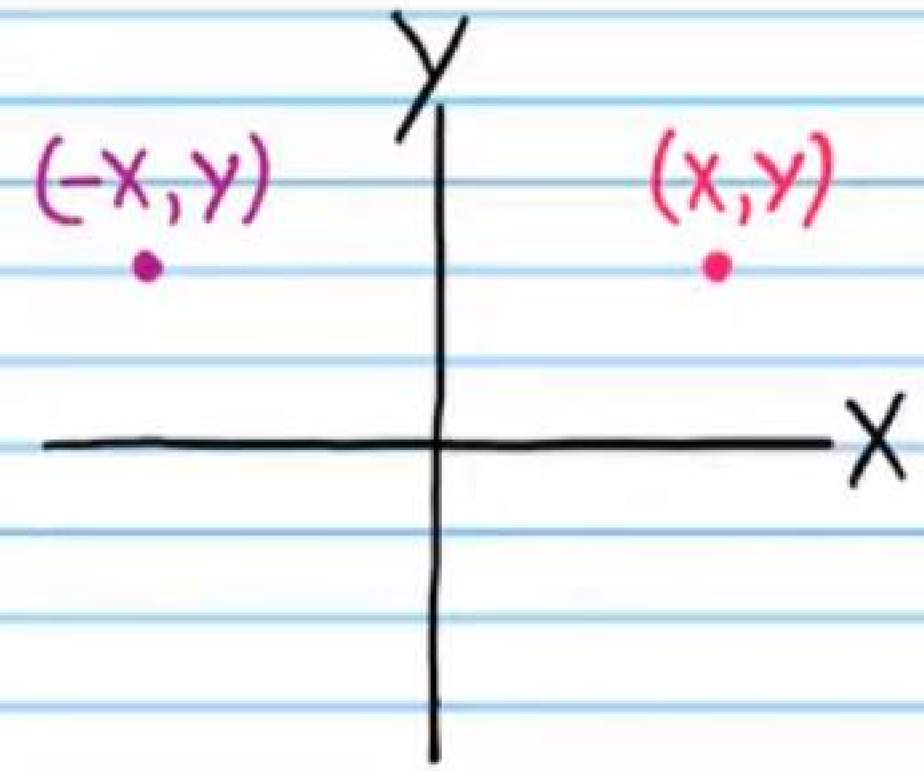
$\boxed{\cot y^4 = 1 + |x|}$

(18) $\frac{1}{y^5} = \ln x \xrightarrow[\text{the x-axis}]{\text{Reflect across}} \frac{1}{(-y)^5} = \ln x$

$\boxed{-\frac{1}{y^5} = \ln x}$

Reflection across the Y-Axis

If (x,y) is a point on the xy-plane, then the reflection of this point across the y-axis is the point with coordinates $(-x,y)$.



Relations are written below in table form. Reflect each point across the y-axis and write the coordinates in the tables at the right.

(19)

X	2	5	-1	0	4
Y	1	3	7	-2	4

Reflect across the y-axis →

X	-2	-5	1	0	-4
Y	1	3	7	-2	4

(20)

X	0	4	17	-3	-5
Y	-1	8	20	-8	-10

Reflect across the y-axis →

X	0	-4	-17	3	5
Y	-1	8	20	-8	-10

(21)

X	-9	4	6	-3	0
Y	-9	-4	10	7	2

Reflect across the y-axis →

X	9	-4	-6	3	0
Y	-9	-4	10	7	2

(22)

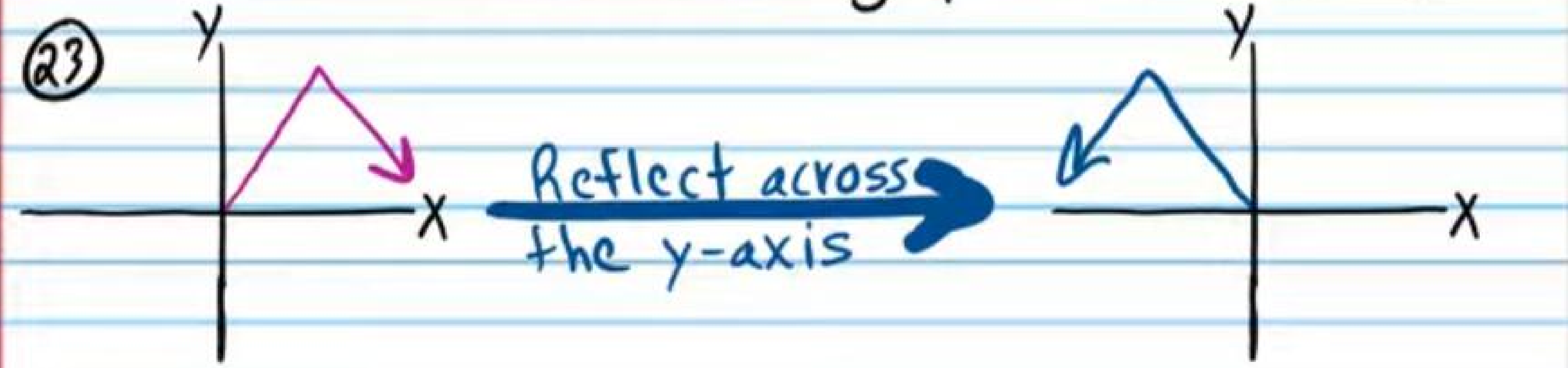
X	-6	4	3	3/4	10
Y	0.1	½	6	0	-8

Reflect across the y-axis →

X	6	-4	-3	-3/4	-10
Y	0.1	½	6	0	-8

Relations are written below in the form of graphs. Draw the reflection of each graph across the y-axis.

(23)



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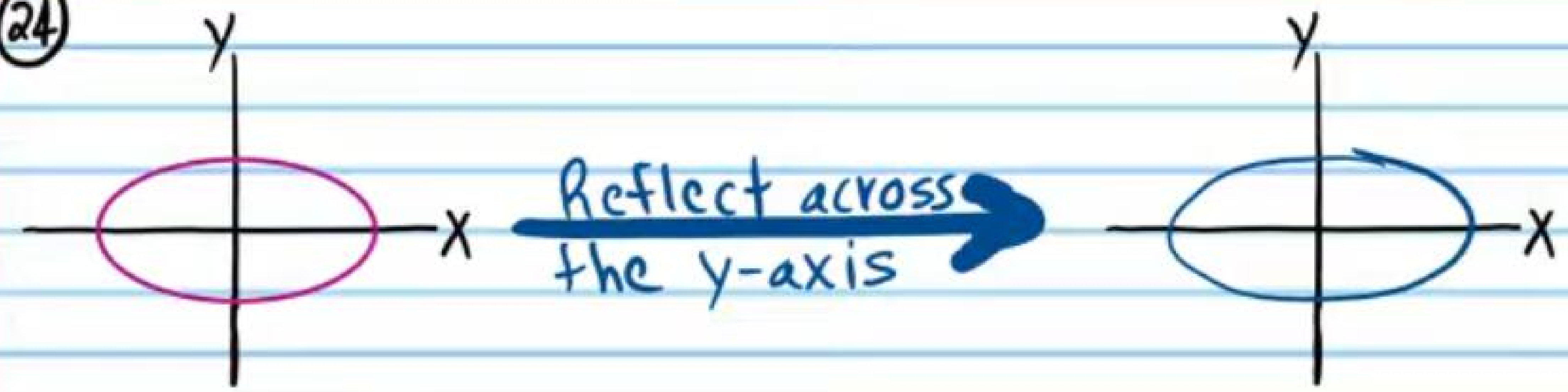

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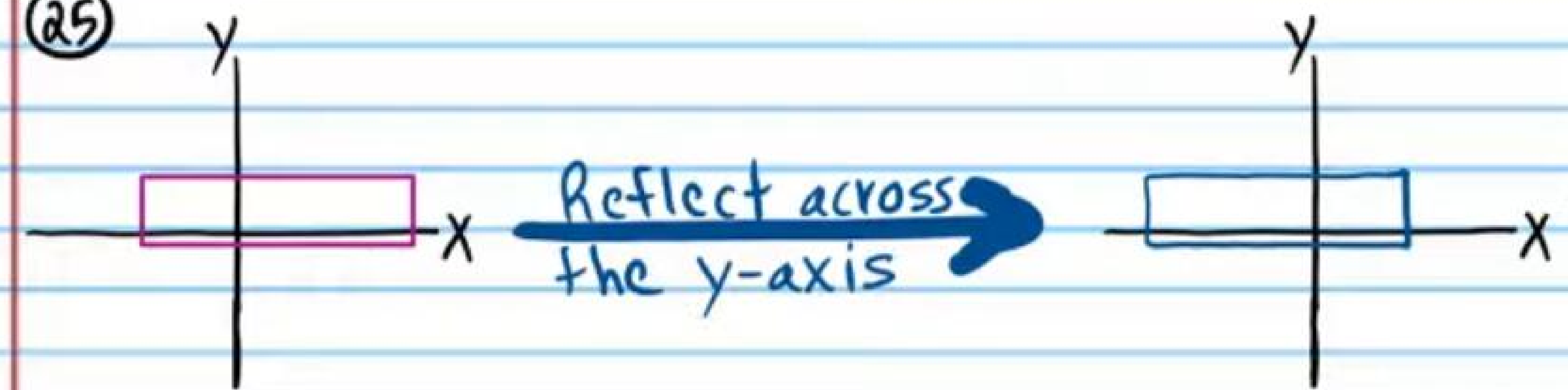

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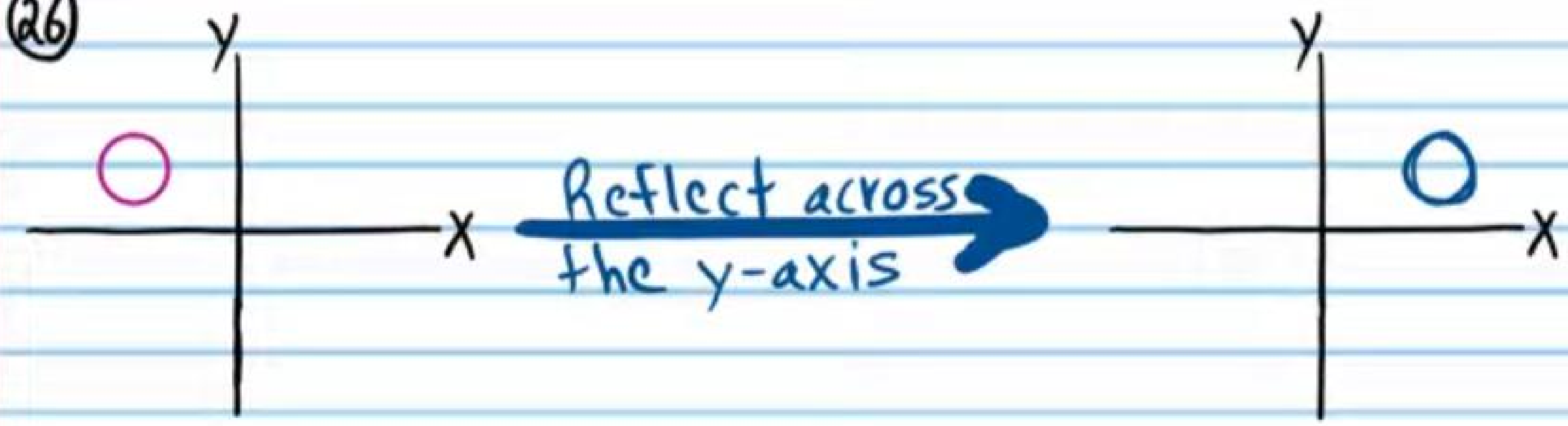

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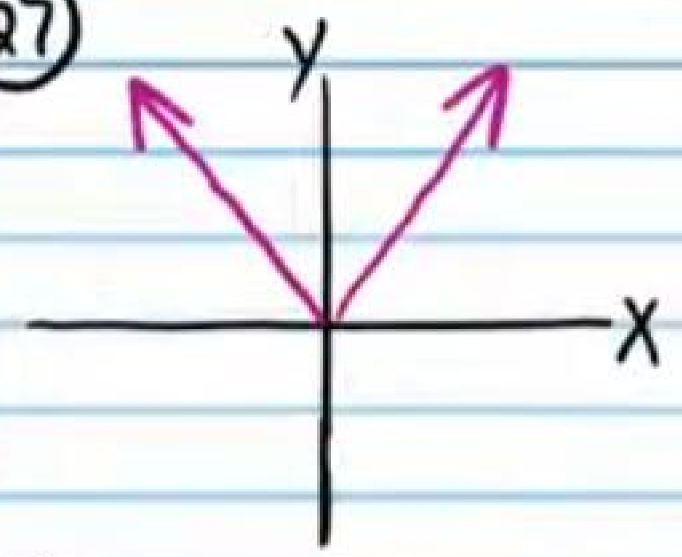
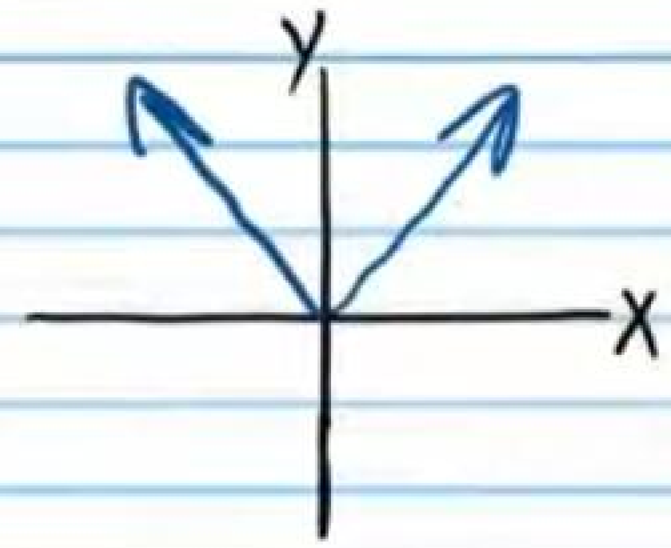


Reflect across the y-axis $\Longrightarrow$



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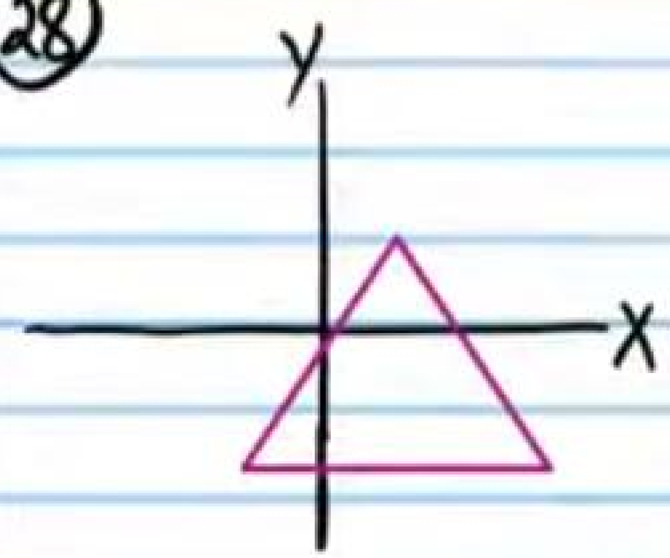
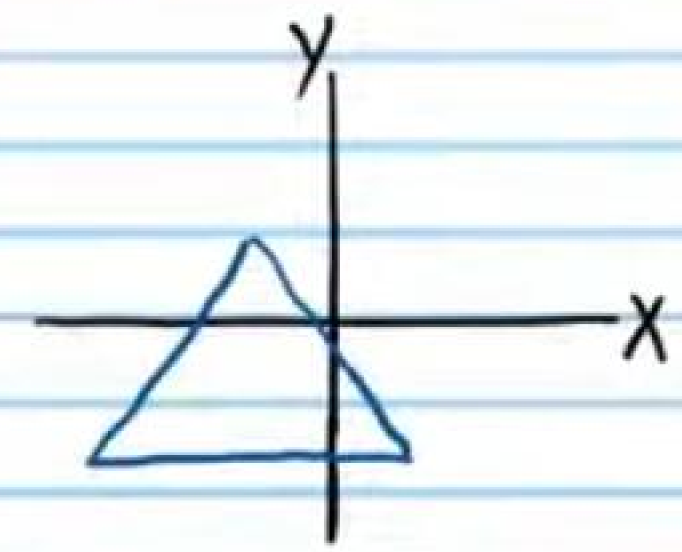


Reflect across the y-axis $\Longrightarrow$



The equations of relations are written below. Write equations of reflections across the y-axis. Simplify if possible.

(29) $x = y - 2$ $\xrightarrow[\text{the y-axis}]{\text{Reflect across}}$ $\boxed{-x = y - 2}$

(30) $y^3 = x^2 + 1$ $\xrightarrow[\text{the y-axis}]{\text{Reflect across}}$ $y^3 = (-x)^2 + 1$

$\boxed{y^3 = x^2 + 1}$

(31) $y^2 = \sin(x+10)$ $\xrightarrow[\text{the y-axis}]{\text{Reflect across}}$ $\boxed{y^2 = \sin(-x+10)}$

(32) $\cot y^4 = 1 + |x|$ $\xrightarrow[\text{the y-axis}]{\text{Reflect across}}$ $\cot y^4 = 1 + |-x|$

$\boxed{\cot y^4 = 1 + |x|}$

(33) $y = x^5$ $\xrightarrow[\text{the y-axis}]{\text{Reflect across}}$ $y = (-x)^5$

$\boxed{y = -x^5}$

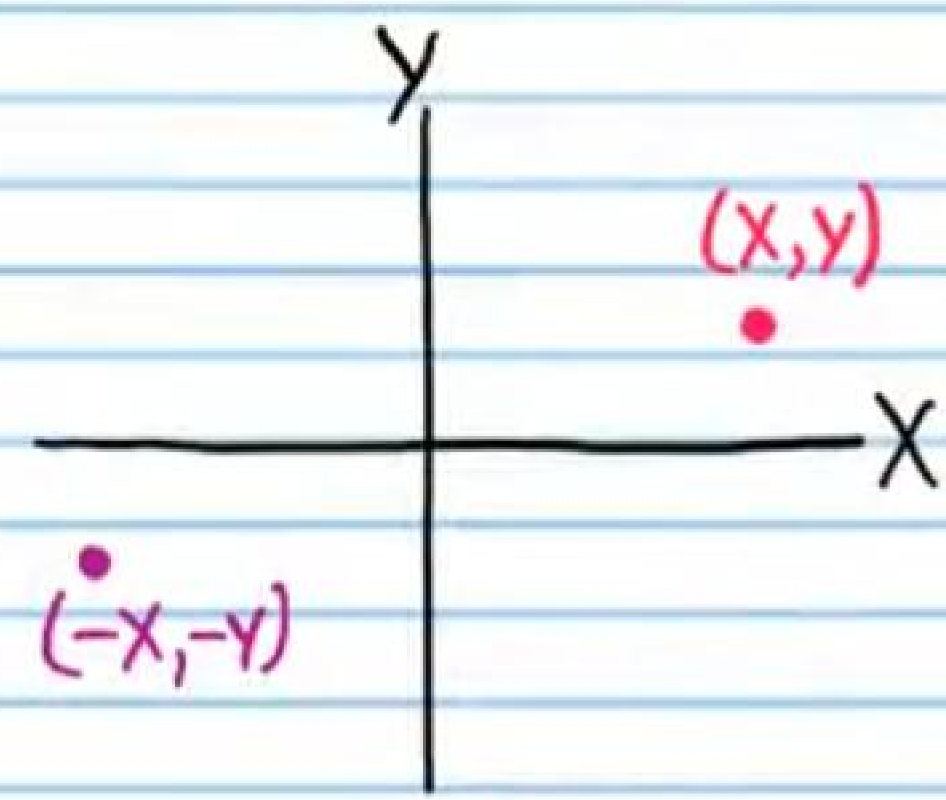
(34) $3^{y-2} = x - 6$ $\xrightarrow[\text{the y-axis}]{\text{Reflect across}}$ $\boxed{3^{y-2} = -x - 6}$

(35) $\log_2 x = \sqrt{y}$ $\xrightarrow[\text{the y-axis}]{\text{Reflect across}}$ $\boxed{\log_2(-x) = \sqrt{y}}$

(36) $\frac{1}{y^5} = \ln x$ $\xrightarrow[\text{the y-axis}]{\text{Reflect across}}$ $\boxed{\frac{1}{y^5} = \ln(-x)}$

Reflection across the Origin

If (x,y) is a point on the xy-plane, then the reflection of this point across the origin is the point with coordinates $(-x,-y)$.



Reflection across the origin can be done in two steps. First, reflect across the x-axis (or y-axis), then reflect across the y-axis (or x-axis).

Relations are written below in table form. Reflect each point across the origin, and write the coordinates in the tables at the right.

(37)

X	2	5	-1	0	4
Y	1	3	7	-2	4

Reflect across the origin →

X	-2	-5	1	0	-4
Y	-1	-3	-7	2	-4

(38)

X	0	4	17	-3	-5
Y	-1	8	20	-8	-10

Reflect across the origin →

X	0	-4	-17	3	5
Y	1	-8	-20	8	10

(39)

X	9	4	6	3	0
Y	-9	-4	10	7	2

Reflect across the origin →

X	-9	-4	-6	-3	0
Y	9	4	-10	-7	-2

(40)

X	-6	4	3	3/4	10
Y	0.1	½	6	0	-8

Reflect across the origin →

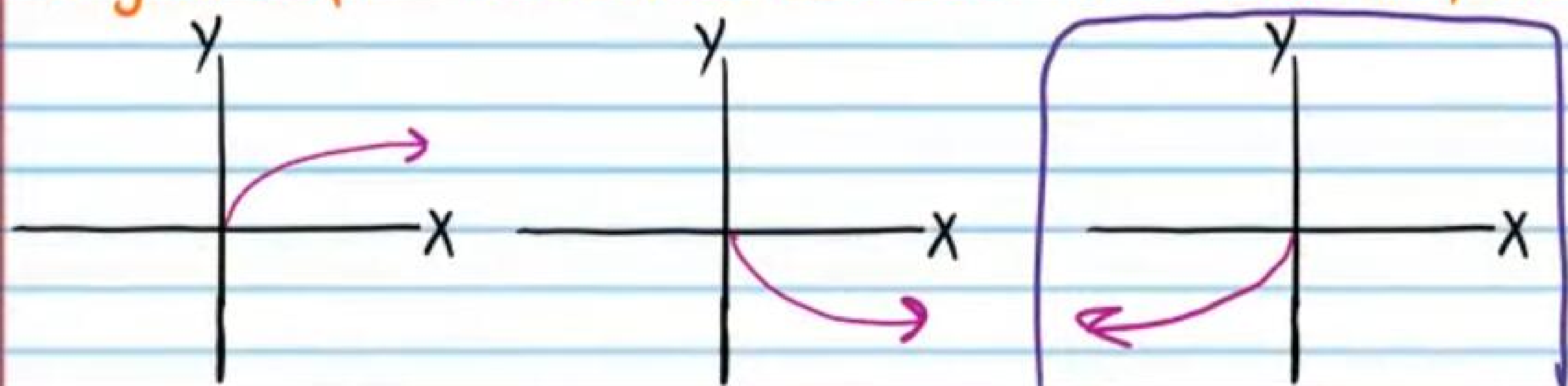
X	6	-4	-3	-3/4	-10
Y	-0.1	$-\frac{1}{2}$	-6	0	8

Relations are written below in the form of graphs. Draw the reflection of each graph across the origin.

(41)

Reflect across the origin →

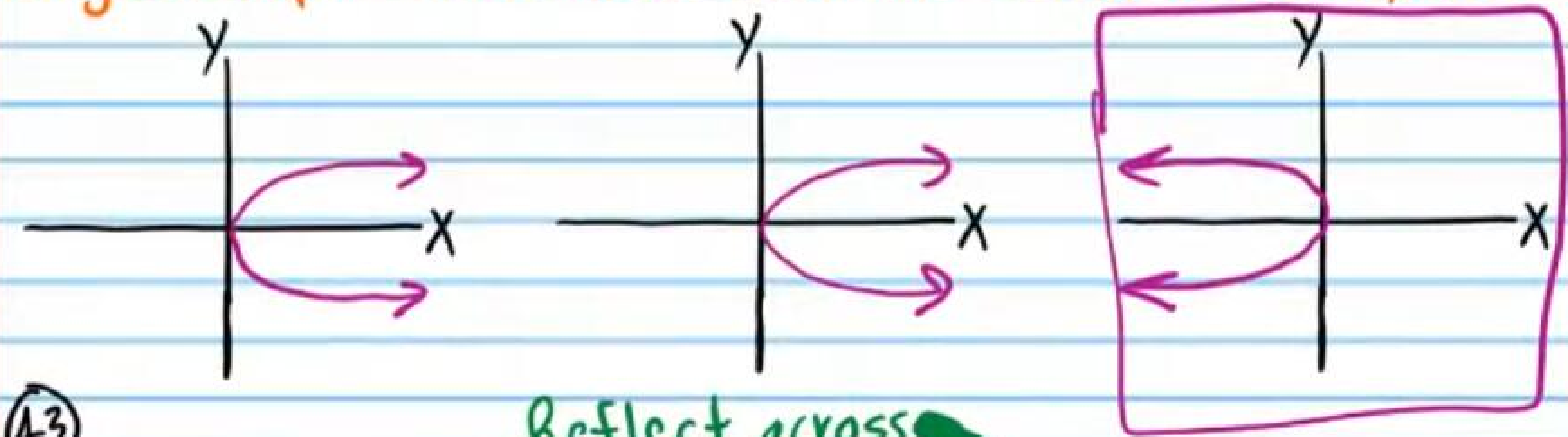
Original Graph | Step 1: Reflect across the x-axis | Step 2: Reflect across the y-axis



(42)

Reflect across the origin →

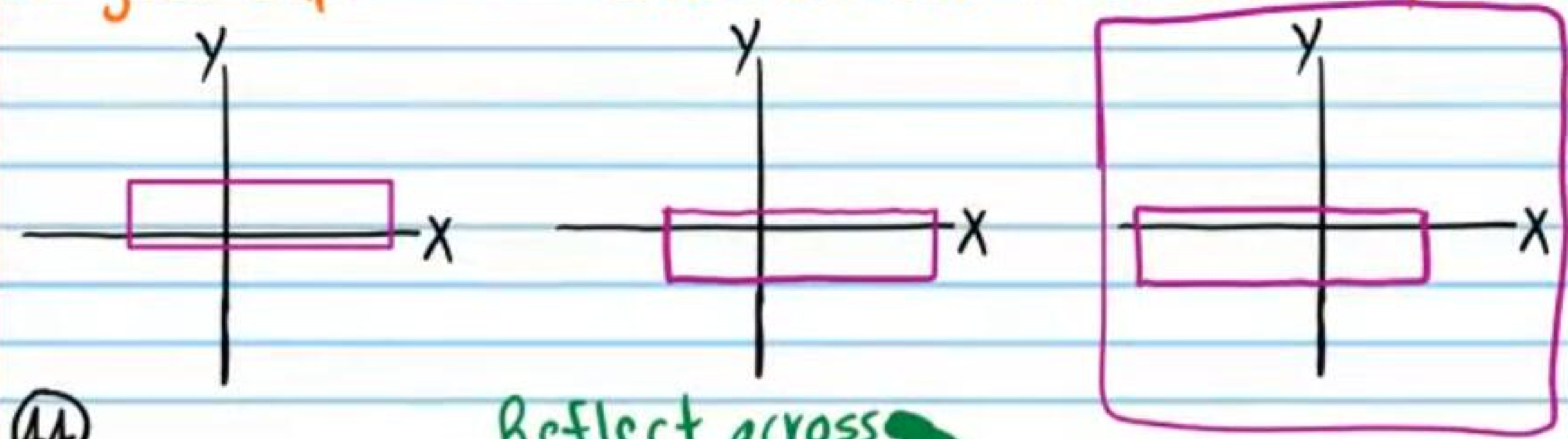
Original Graph | Step 1: Reflect across the x-axis | Step 2: Reflect across the y-axis



(43)

Reflect across the origin →

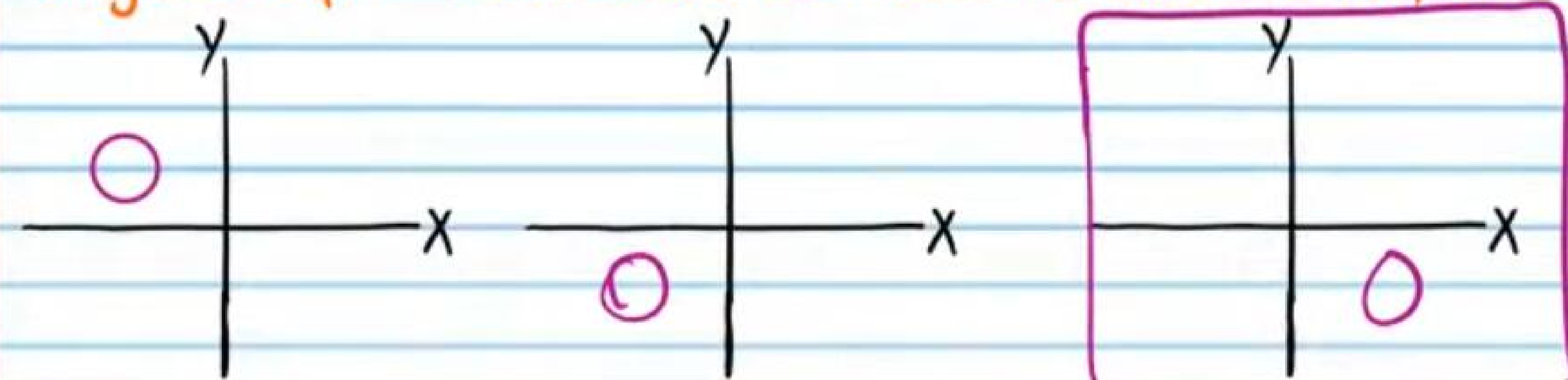
Original Graph | Step 1: Reflect across the x-axis | Step 2: Reflect across the y-axis



(44)

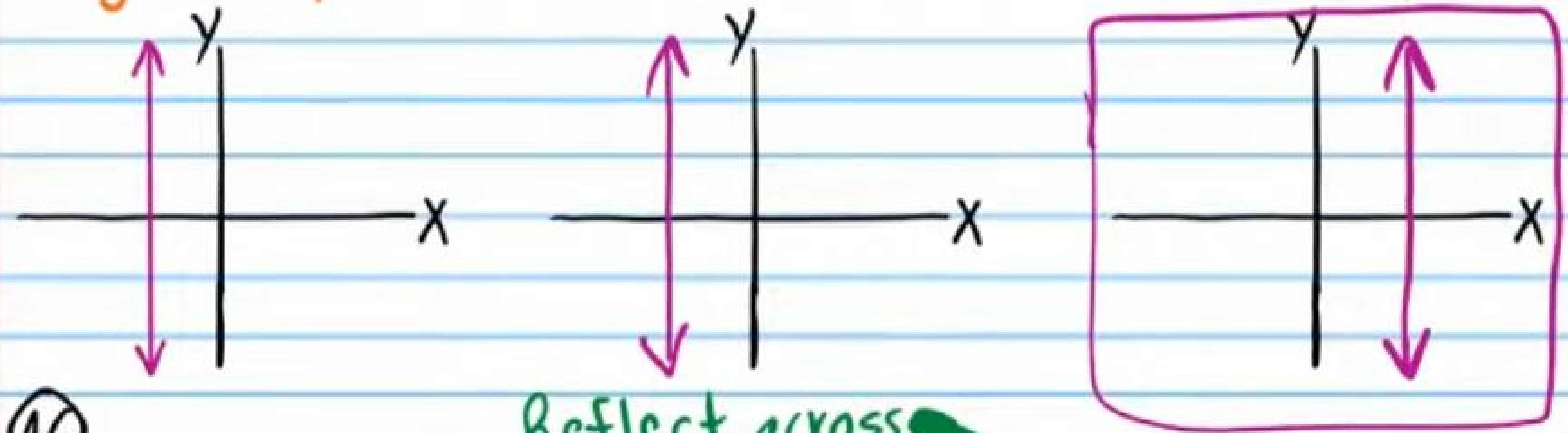
Reflect across the origin →

Original Graph | Step 1: Reflect across the x-axis | Step 2: Reflect across the y-axis



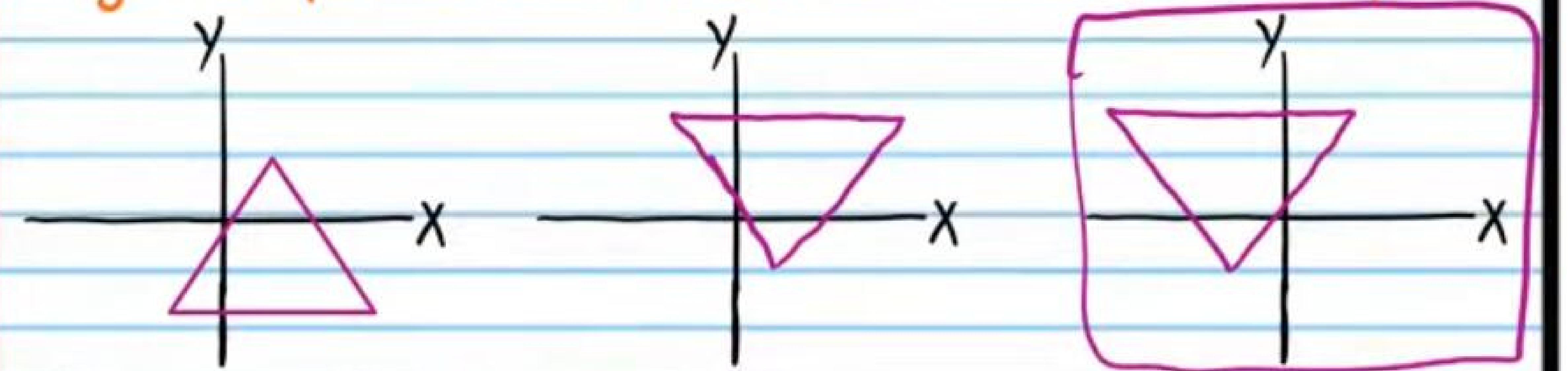
(45) $\xrightarrow{\text{Reflect across the origin}}$

Original Graph	Step 1 Reflect across the x-axis	Step 2 Reflect across the y-axis



(46) $\xrightarrow{\text{Reflect across the origin}}$

Original Graph	Step 1 Reflect across the x-axis	Step 2 Reflect across the y-axis



The equations of relations are written below. Write equations of reflections across the origin. Simplify if possible.

(47) $x = y - 2 \xrightarrow{\text{Reflect across the origin}} -x = -y - 2$

$$\boxed{x = y + 2}$$

(48) $y^3 = x^2 + 1 \xrightarrow{\text{Reflect across the origin}} (-y)^3 = (-x)^2 + 1$

$$\boxed{-y^3 = x^2 + 1}$$

(49) $y^2 = \sin(x+10) \xrightarrow{\text{Reflect across the origin}} (-y)^2 = \sin(-x+10)$

$$\boxed{y^2 = \sin(-x+10)}$$

(50) $\log_2 x = \sqrt{y} \xrightarrow{\text{Reflect across the origin}} \boxed{\log_2(-x) = \sqrt{-y}}$

(51) $y = x^5 \xrightarrow{\text{Reflect across the origin}} -y = (-x)^5$

$$-y = -x^5$$

$$\boxed{y = x^5}$$

(52) $3^{y-2} = x - 6 \xrightarrow{\text{Reflect across the origin}} \boxed{3^{-y-2} = -x - 6}$

(53) $\cot y^4 = 1 + |x| \xrightarrow{\text{Reflect across the origin}} \cot(-y)^4 = 1 + |-x|$

$$\boxed{\cot y^4 = 1 + |x|}$$

(54) $\frac{1}{y^9} = \ln x \xrightarrow{\text{Reflect across the origin}} \frac{1}{(-y)^9} = \ln(-x)$

$$\boxed{-\frac{1}{y^9} = \ln(-x)}$$

Opposite Angle Identities

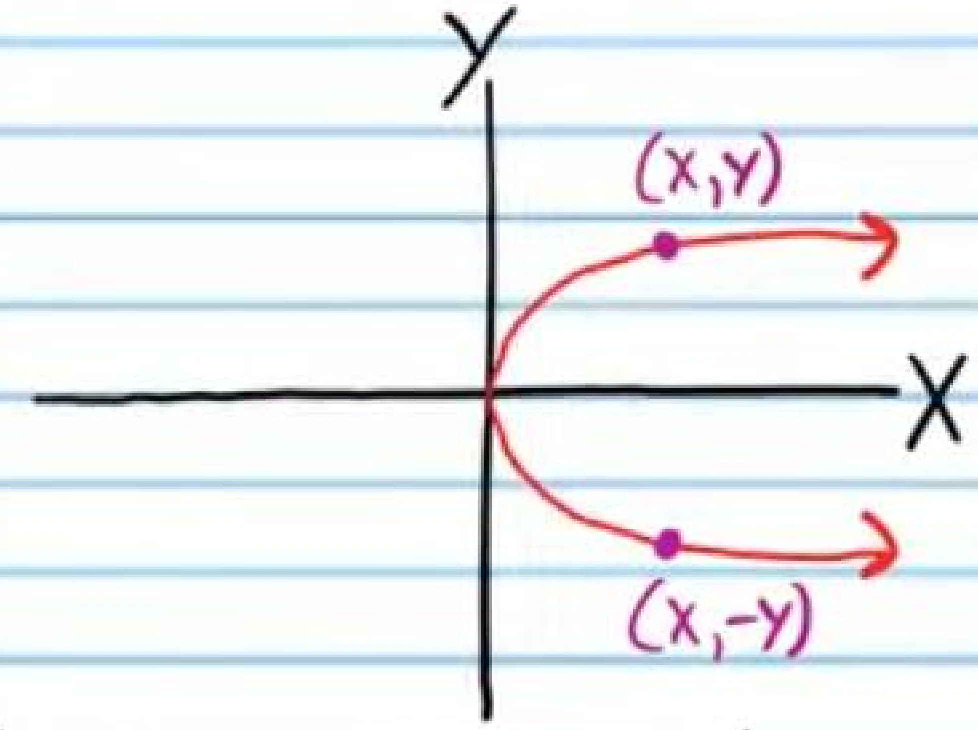
$\sin(-\theta) = -\sin\theta$	$\cos(-\theta) = \cos\theta$	$\tan(-\theta) = -\tan\theta$
$\csc(-\theta) = -\csc\theta$	$\sec(-\theta) = \sec\theta$	$\cot(-\theta) = -\cot\theta$

Symmetry

Symmetry about the X-Axis

A graph is symmetric about the x-axis if for every point (x, y) on the graph, there exists a point $(x, -y)$ that is also on the graph.

Example I



Each table below has points collected from a graph that is symmetric about the x-axis. Fill the empty spaces with other points from each corresponding graph.

55

X	6	7	8	1	-2	6	7	8	1	-2
Y	2	8	-2	3	4	-2	-8	2	-3	-4

56

X	0	-3	2	4	-10	0	-3	2	4	-10
Y	8	-9	3	0	0.5	-8	9	-3	0	-0.5

57

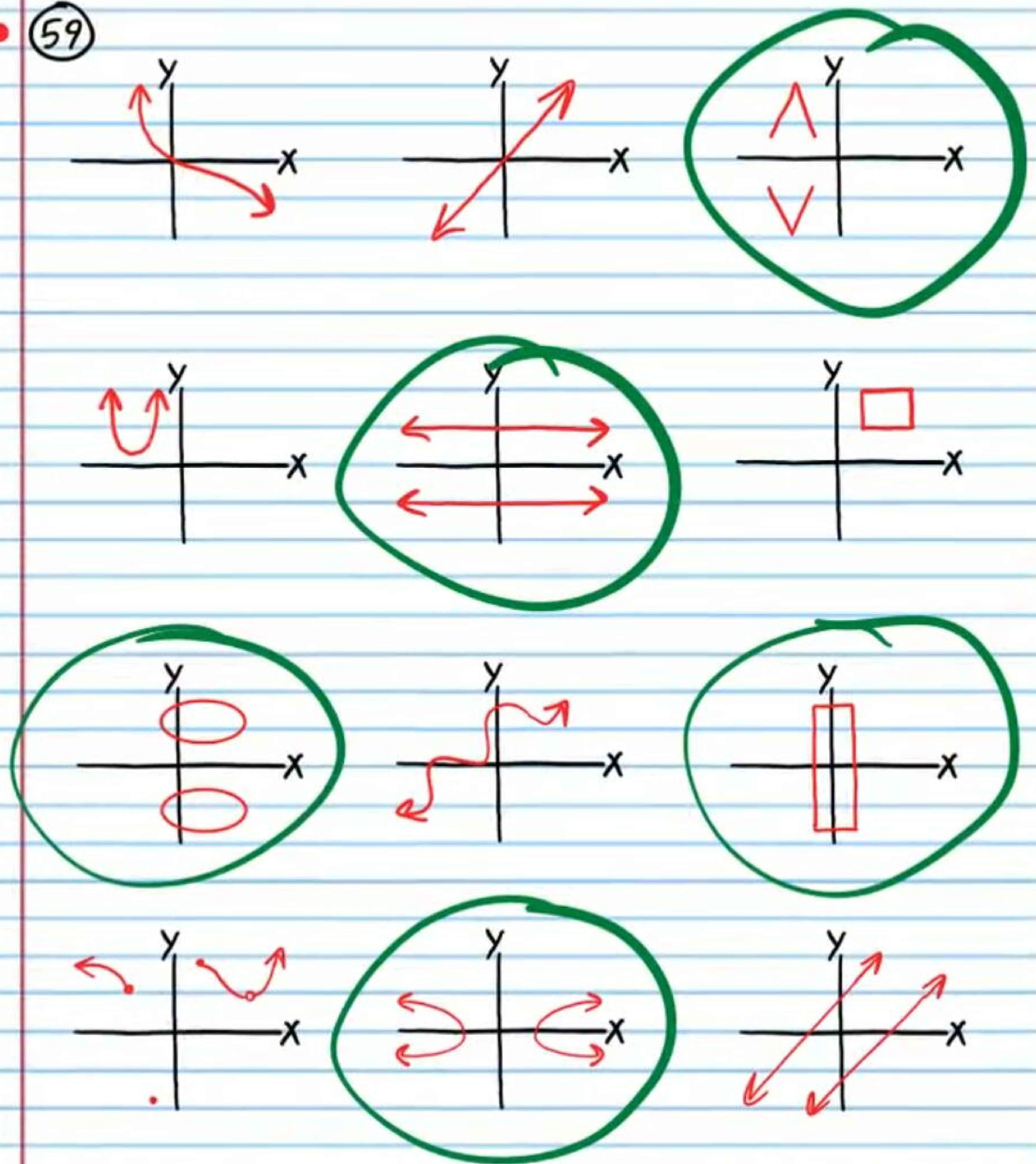
X	0	3	-5	1.6	-2	0	3	-5	1.6	-2
Y	-1	4	-5	2.4	-1	1	-4	5	-2.4	1

58

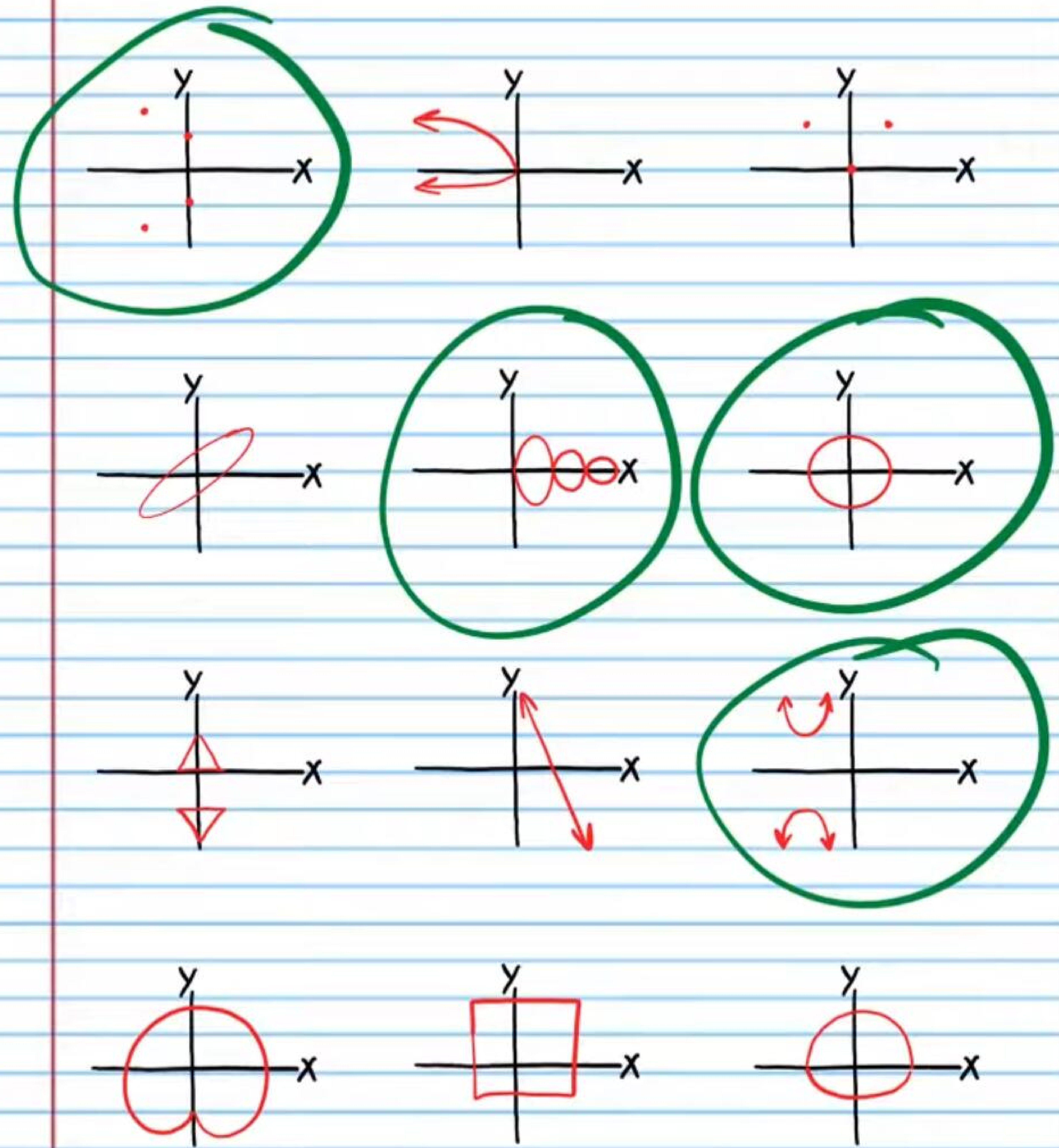
X	2	-3	5	6	-1	2	-3	5	6	-1
Y	7	8	10	-9	3	-7	-8	-10	9	-3

Circle the graphs that appear to be symmetric about the x-axis.

59



(60)



Determining if an Equation Will Produce a Graph That Is Symmetric about the X-Axis

To determine if the graph of an equation is symmetric about the x-axis, replace y with $-y$ in the equation and simplify. If the simplified equation is the same as the original equation, then the graph of the original equation is symmetric about the x-axis.

Example I: $y^2 = x$

Replace y with $-y$ $\Longrightarrow$ $(-y)^2 = x$

Simplify if possible $\Longrightarrow$ $y^2 = x$

The same

Therefore, the graph of $y^2 = x$ is symmetric about the x-axis.

Example II: $y = x^3 + 2$

Replace y with $-y$ $\Longrightarrow$ $-y = x^3 + 2$

Not possible to simplify $\Longrightarrow$ $-y = x^3 + 2$

Not the same

Therefore, the graph of $y = x^3 + 2$ is not symmetric about the x-axis.

Determine if the graph of each relation below is symmetric about the x-axis.

(61) $x = |y| - 3$

$x = |-y| - 3$

$x = |y| - 3$

✓

(62) $-y = 2 + x^4$

$-(-y) = 2 + x^4$

$y = 2 + x^4$

✗

(63) $x^2 + 2x + 1 = \sec y$

$x^2 + 2x + 1 = \sec(-y)$

$x^2 + 2x + 1 = \sec y$

✓

(64) $\frac{1}{x} = y^2 - y^3$

$\frac{1}{x} = (-y)^2 - (-y)^3$

$\frac{1}{x} = y^2 - (-y^3)$

$\frac{1}{x} = y^2 + y^3$

✗

(65) $4x - 2y = y^7$

$4x - 2(-y) = (-y)^7$

$4x + 2y = -y^7$

$-1(4x + 2y) = (-y^7)(-1)$

$-4x - 2y = y^7$

✗

(66) $5^{y+|y|} - 4 = \csc x$

$5^{-y+|-y|} - 4 = \csc x$

$5^{-y+|y|} - 4 = \csc x$

✗

(67) $x^2 y = \frac{1}{y}$

$x^2(-y) = \frac{1}{-y}$

$-x^2 y = -\frac{1}{y}$

$-1(-x^2 y) = \left(-\frac{1}{y}\right)(-1)$

$x^2 y = \frac{1}{y}$

✓

(68) $\tan y = \sin y + xy$

$\tan(-y) = \sin(-y) + x(-y)$

$-\tan y = -\sin y - xy$

$-1(-\tan y) = (-\sin y - xy)(-1)$

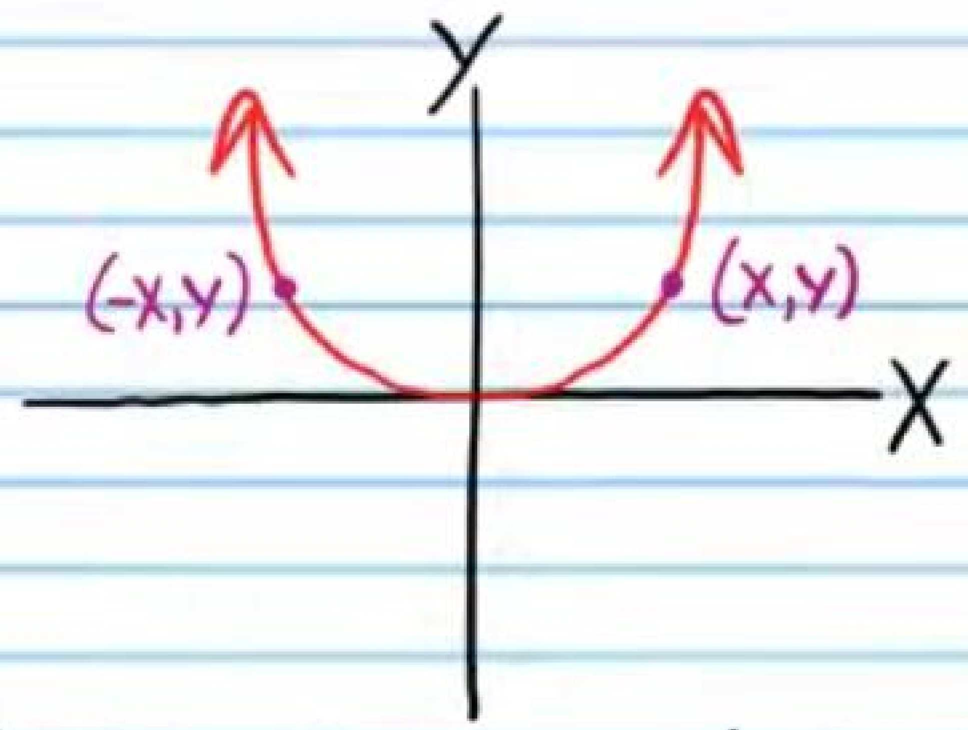
$\tan y = \sin y + xy$

✓

Symmetry about the Y-Axis

A graph is symmetric about the y-axis if for every point (x,y) on the graph, there exists a point (-x,y) that is also on the graph.

Example III:



Each table below has points collected from a graph that is symmetric about the y-axis. Fill the empty spaces with other points from each corresponding graph.

(69)

X	6	7	8	1	-2	-6	-7	-8	-1	2
Y	2	8	-2	3	4	2	8	-2	3	4

(70)

X	0	-3	2	4	-10	0	3	-2	-4	10
Y	8	-9	3	0	0.5	8	-9	3	0	0.5

(71)

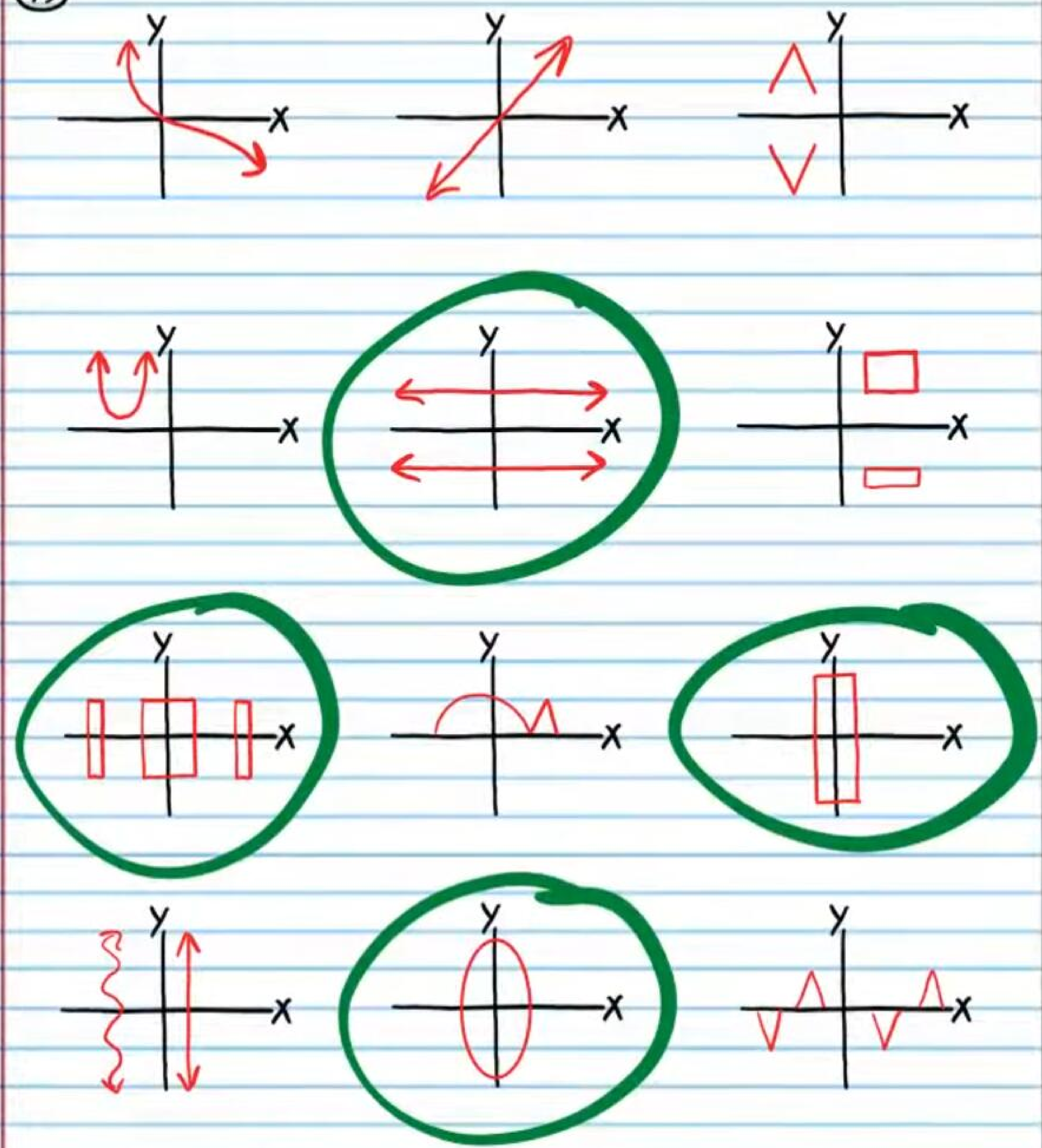
X	0	3	-5	1.6	-2	0	-3	5	-1.6	2
Y	-1	4	-5	2.4	-1	-1	4	-5	2.4	-1

(72)

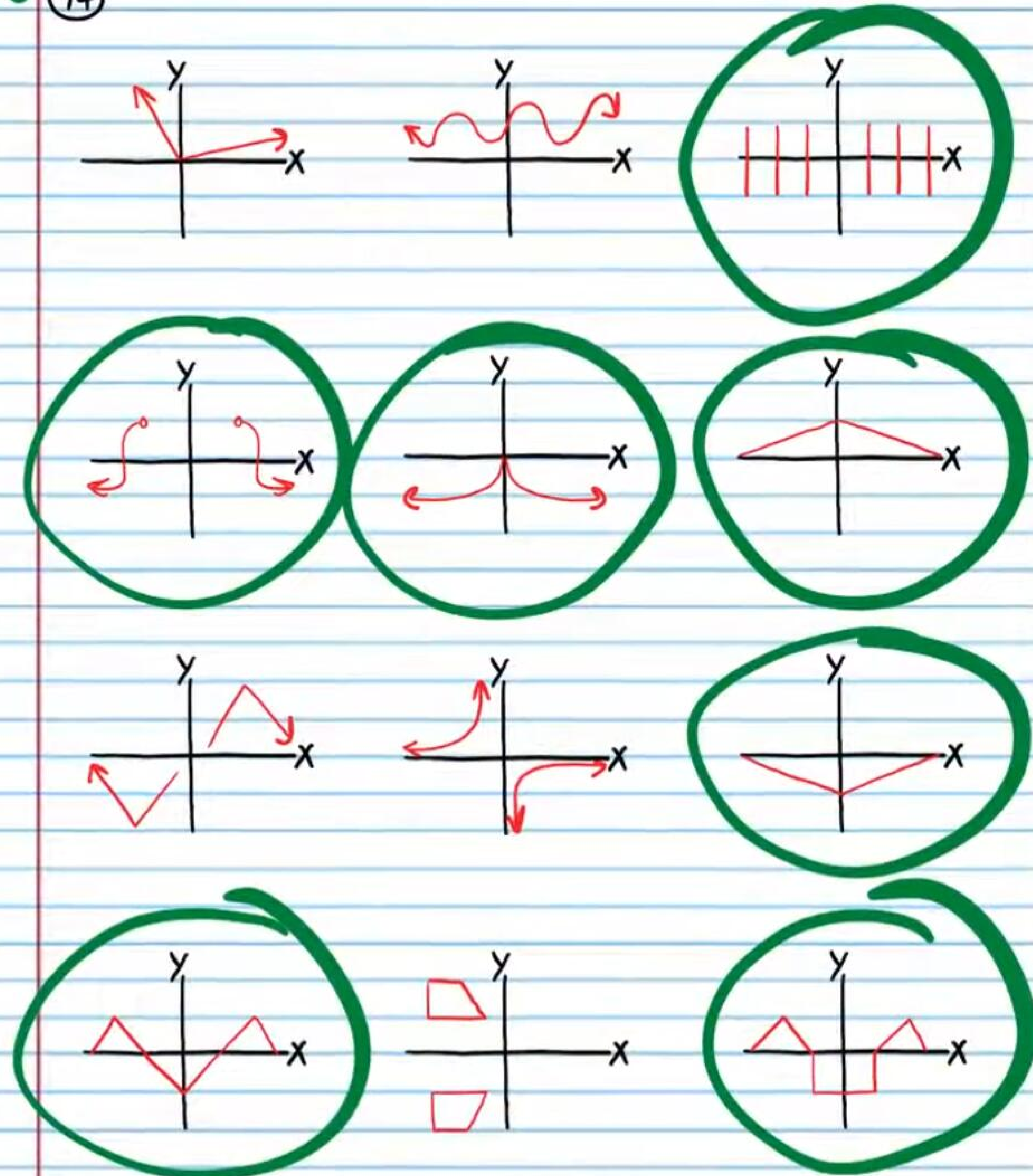
X	2	-3	5	6	-1	-2	3	-5	-6	1
Y	7	8	10	-9	3	7	8	10	-9	3

Circle the graphs that appear to be symmetric about the y-axis.

(73)



(74)



Determining if an Equation Will Produce a Graph That Is Symmetric about the Y-Axis

To determine if the graph of an equation is symmetric about the y-axis, replace x with $-x$ in the equation and simplify. If the simplified equation is the same as the original equation, then the graph of the original equation is symmetric about the y-axis.

Example III: $y = x^4 + x^2$ $\quad (x, y) \longrightarrow (-x, y)$

Replace x with $-x$ $\Longrightarrow$ $y = (-x)^4 + (-x)^2$

Simplify if possible $\Longrightarrow$ $y = x^4 + x^2$

The same

Therefore, the graph of $y = x^4 + x^2$ is symmetric about the y-axis.

Example IV: $y = x^3 + x^2$

Replace x with $-x$ $\Longrightarrow$ $y = (-x)^3 + (-x)^2$

Simplify if possible $\Longrightarrow$ $y = -x^3 + x^2$

Not the same

Therefore, the graph of $y = x^3 + x^2$ is not symmetric about the y-axis.

Determine if the graph of each relation below is symmetric about the y-axis.

(75) $x = |y| - 3$

$-x = |y| - 3$

$-1(-x) = (|y| - 3)(-1)$

$x = -|y| + 3$

✗

(76) $-y = 2 + x^4$

$-y = 2 + (-x)^4$

$-y = 2 + x^4$

✓

(77) $x^2 + 2x + 1 = \sec y$

$(-x)^2 + 2(-x) + 1 = \sec y$

$x^2 - 2x + 1 = \sec y$

✗

(78) $y = \sin^2 x$

$y = \sin^2(-x)$

$y = [\sin(-x)]^2$

$y = [-\sin x]^2$

$y = [\sin x]^2$

$y = \sin^2 x$

✓

(79) $4x - 2y = y^7$

$4(-x) - 2y = y^7$

$-4x - 2y = y^7$

✗

(80) $5^{y^2 + |y|} - 4 = \csc x$

$5^{y^2 + |y|} - 4 = \csc(-x)$

$5^{y^2 + |y|} - 4 = -\csc x$

$-1(5^{y^2 + |y|} - 4) = (-\csc x)(-1)$

$-5^{y^2 + |y|} + 4 = \csc x$

✗

(81) $x^2 y = \frac{1}{y}$

$(-x)^2 y = \frac{1}{y}$

$x^2 y = \frac{1}{y}$

✓

(82) $x^6 + 3x^4 - 2x^2 = y + 1$

$(-x)^6 + 3(-x)^4 - 2(-x)^2 = y + 1$

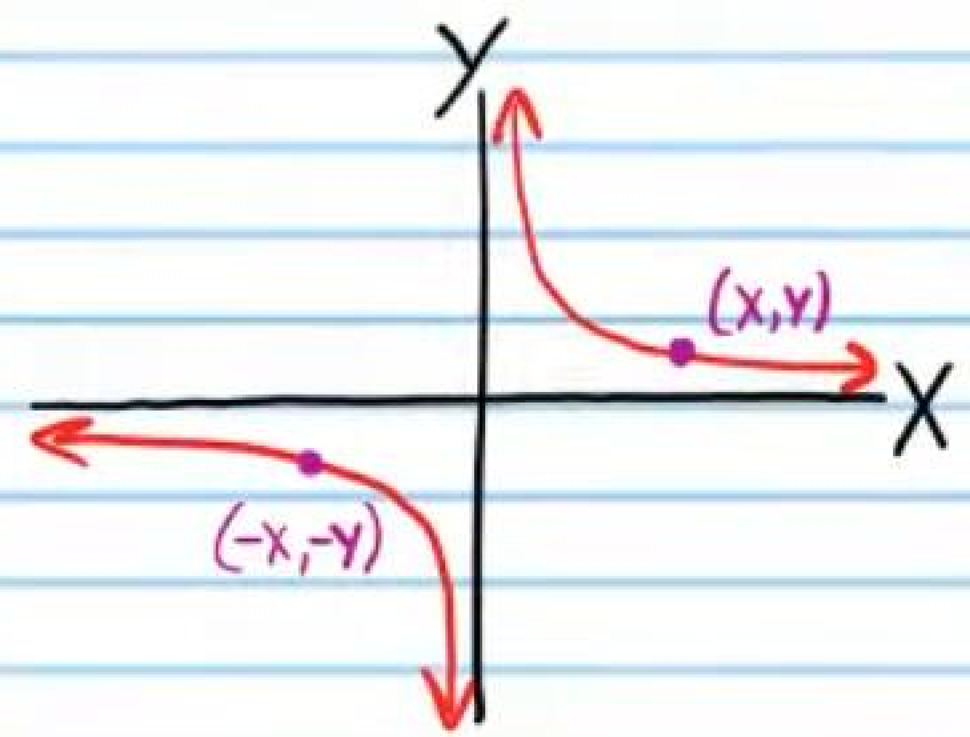
$x^6 + 3x^4 - 2x^2 = y + 1$

✓

Symmetry about the Origin

A graph is symmetric about the origin if for every point (x,y) on the graph, there exists a point (-x,-y) that is also on the graph.

Example I:



Each table below has points collected from a graph that is symmetric about the origin. Fill the empty spaces with other points from each corresponding graph.

(83)

x	6	7	8	1	-2	-6	-7	-8	-1	2
y	2	8	-2	3	4	-2	-8	2	-3	-4

(84)

x	0	-3	2	4	-10	0	3	-2	-4	10
y	8	-9	3	0	0.5	-8	9	-3	0	-0.5

(85)

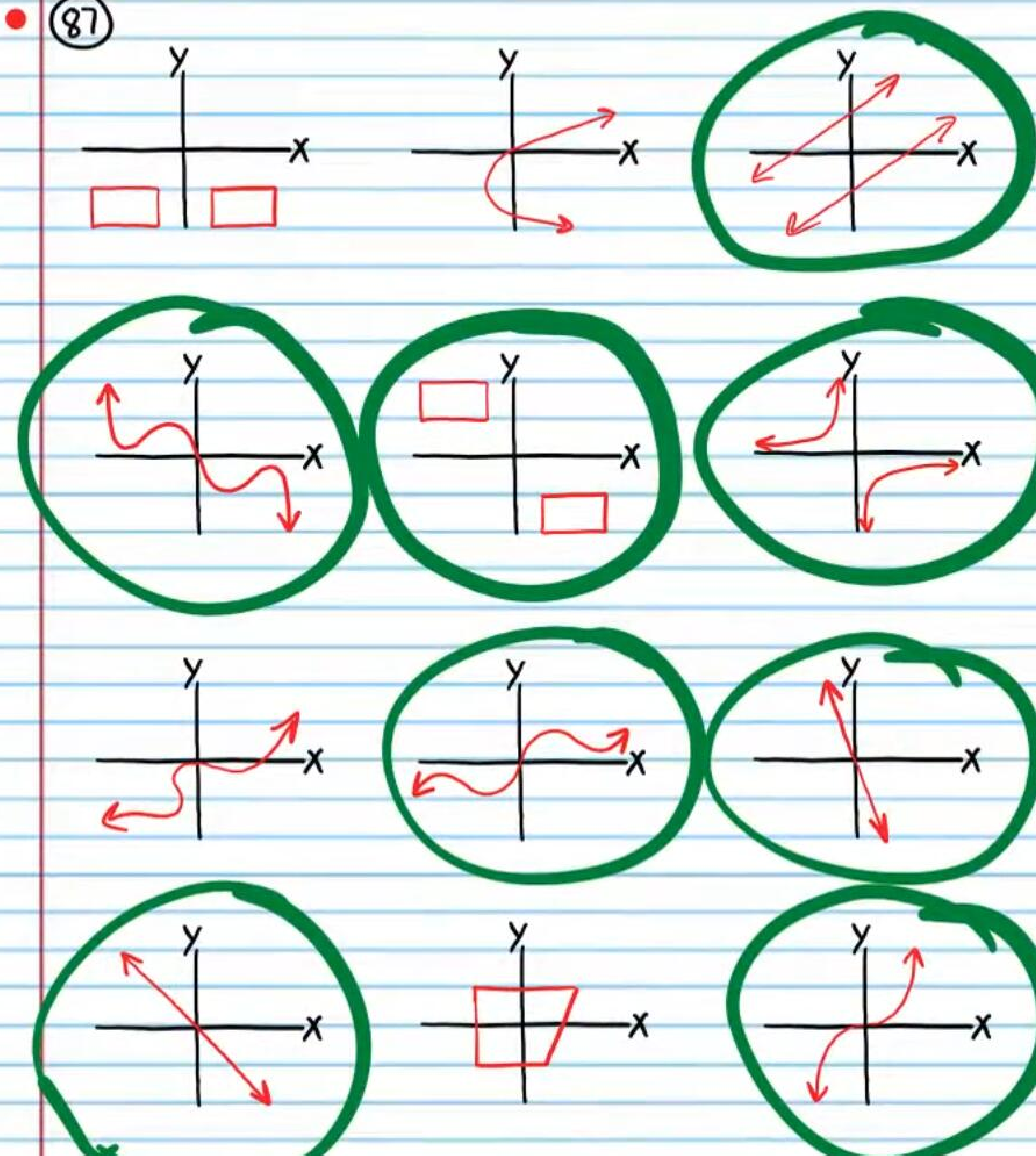
x	0	3	-5	1.6	-2	0	-3	5	-1.6	2
y	-1	4	-5	2.4	-1	1	-4	5	-2.4	1

(86)

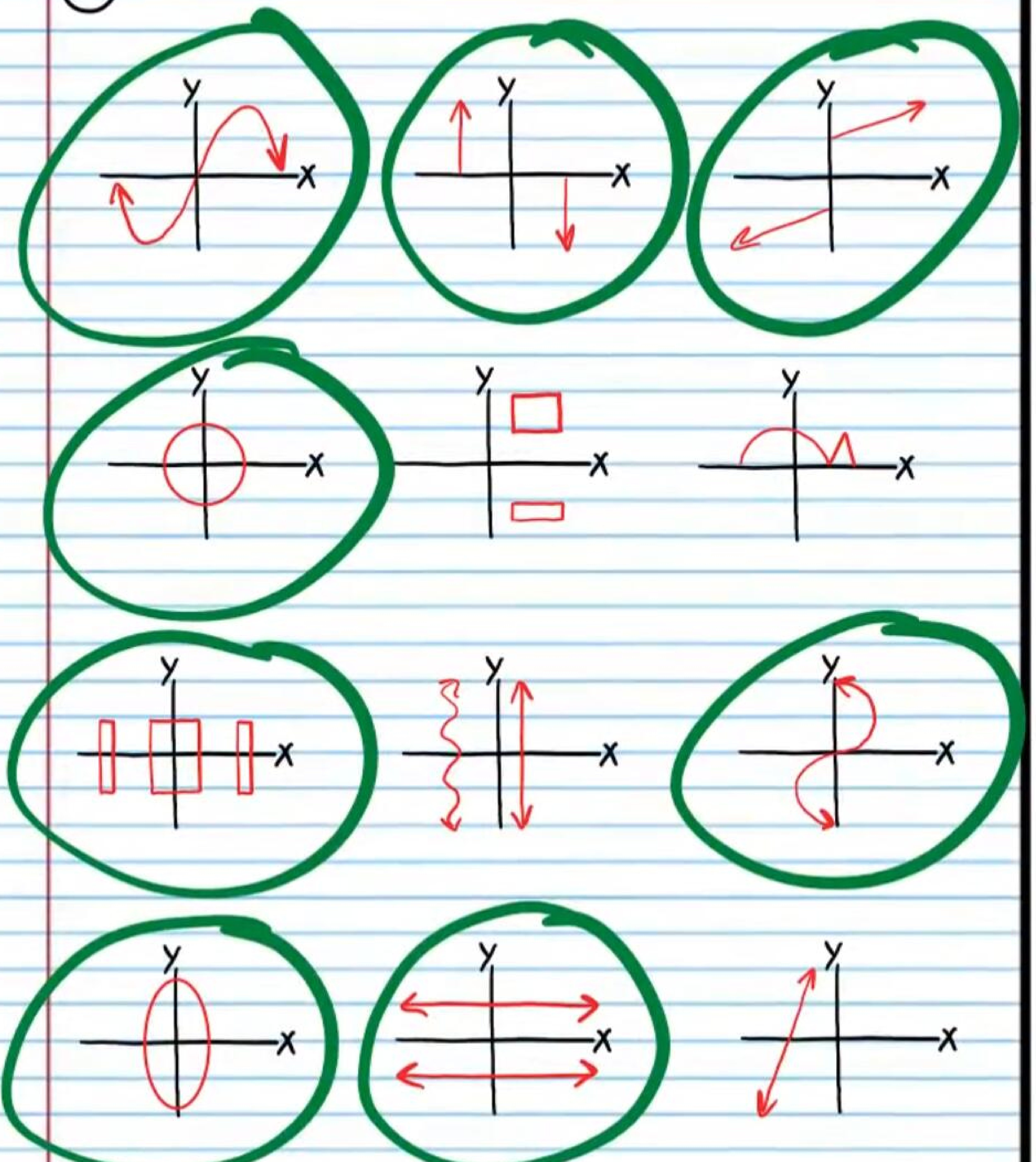
x	2	-3	5	6	-1	-2	3	-5	-6	1
y	7	8	10	-9	3	-7	-8	-10	9	-3

Circle the graphs that appear to be symmetric about the origin.

(87)



88



Determining if an Equation Will Produce a Graph That Is Symmetric about the Origin

To determine if the graph of an equation is symmetric about the origin, replace x with $-x$ and y with $-y$ in the equation and simplify. If the simplified equation is the same as the original equation, then the graph of the original equation is symmetric about the origin.

Example V: $y = \frac{1}{x}$

Replace x with $-x$ and y with $-y$. $\Longrightarrow$ $-y = \frac{1}{-x}$

Simplify if possible $\Longrightarrow$ $y = \frac{1}{x}$

The same

Therefore the graph of $y = \frac{1}{x}$ is symmetric about the origin.

Example VI: $y = |x|$

Replace x with $-x$ and y with $-y$. $\Longrightarrow$ $-y = |-x|$

Simplify if possible $\Longrightarrow$ $-y = |x|$

Not the same

Therefore, the graph of $y = |x|$ is not symmetric about the origin.

Determine if the graph of each relation below is symmetric about the origin.

(89) $x^2 - y^2 = 5$

$$(-x)^2 - (-y)^2 = 5$$

$$x^2 - y^2 = 5$$

(90) $x^2 + 2x + 1 = \sec y$

$$(-x)^2 + 2(-x) + 1 = \sec(-y)$$

$$x^2 - 2x + 1 = \sec y$$

(91) $|x| - 2 = |y|$

$$|-x| - 2 = |-y|$$

$$|x| - 2 = |y|$$

(92) $y = x^3 - \sin x$

$$-y = (-x)^3 - \sin(-x)$$

$$-y = -x^3 - (-\sin x)$$

$$-y = -x^3 + \sin x$$

$$-1(-y) = (-x^3 + \sin x)(-1)$$

$$y = x^3 - \sin x$$

(93) $4x - 2y = y^7$

$$4(-x) - 2(-y) = (-y)^7$$

$$-4x + 2y = -y^7$$

$$-1(-4x + 2y) = (-y^7)(-1)$$

$$4x - 2y = y^7$$

(94) $5^{y^2 + |y|} - 4 = \csc x$

$$5^{(-y)^2 + |-y|} - 4 = \csc(-x)$$

$$5^{y^2 + |y|} - 4 = -\csc x$$

(95) $x^2 y = \frac{1}{y}$

$$(-x)^2(-y) = \frac{1}{-y}$$

$$x^2(-y) = -\frac{1}{y}$$

$$-x^2 y = -\frac{1}{y}$$

$$-1(-x^2 y) = \left(-\frac{1}{y}\right)(-1)$$

$$x^2 y = \frac{1}{y}$$

(96) $-y = 2 + x^4$

$$-(-y) = 2 + (-x)^4$$

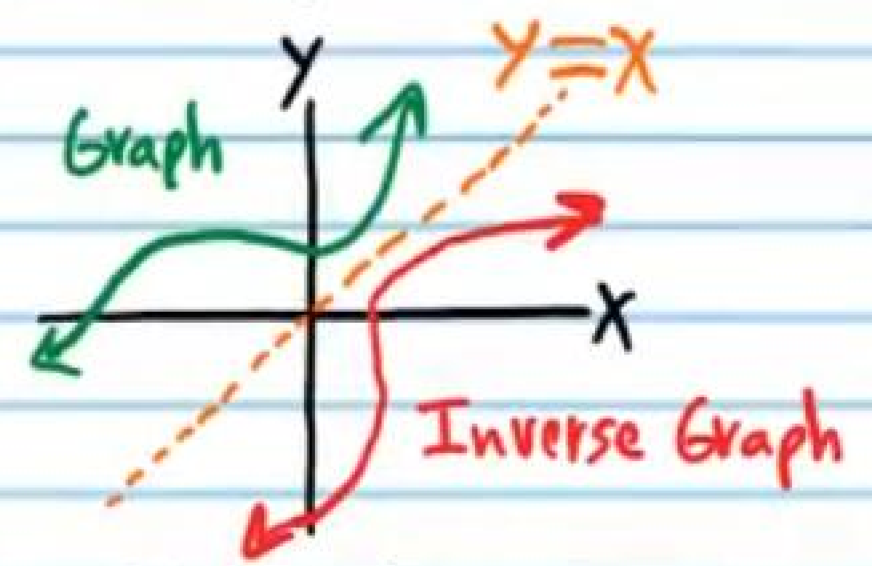
$$y = 2 + x^4$$

Inversion

Two relations are inverses if for every ordered pair (x,y) pertaining to one relation, there exists an ordered pair (y,x) pertaining to the other relation.

The graph of an inverse relation is obtained by reflecting the original graph across the line $y=x$.

Example VII:



Relations are written below in table form. Write the coordinates of the corresponding points of the inverse relations (this is reflection across the line $y=x$).

(97)

X	2	5	-1	0	4
Y	1	3	7	-2	4

Write the inverse relation i.e. Reflection across $y=x$ →

X	1	3	7	-2	4
Y	2	5	-1	0	4

(98)

X	0	4	17	-3	-5
Y	-1	8	20	-8	-10

Write the inverse relation i.e. Reflection across $y=x$ →

X	-1	8	20	-8	-10
Y	0	4	17	-3	-5

(99)

X	9	4	6	3	0
Y	-9	-4	10	7	2

Write the inverse relation i.e. Reflection across $y=x$ →

X	-9	-4	10	7	2
Y	9	4	6	3	0

(100)

X	-6	4	3	3/4	10
Y	0.1	1/2	6	0	-8

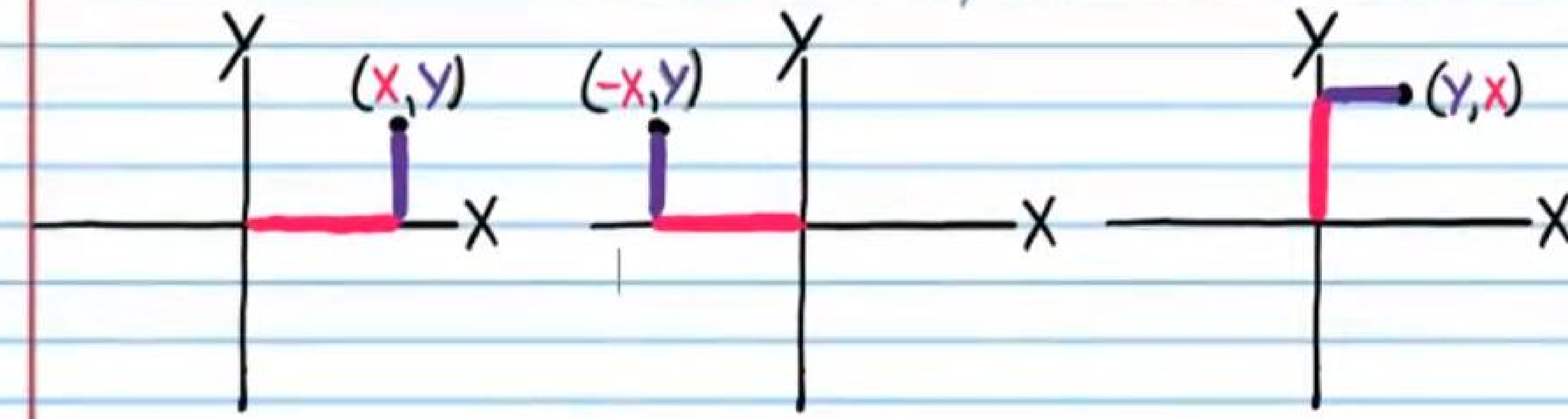
Write the inverse relation i.e. Reflection across $y=x$ →

X	0.1	1/2	6	0	-8
Y	-6	4	3	3/4	10

However, it is easier to draw inverse graphs by transforming the original graph in two steps.

Original Graph — Step 1 — Step 2

Reflect across the y-axis — Rotate 90° clockwise

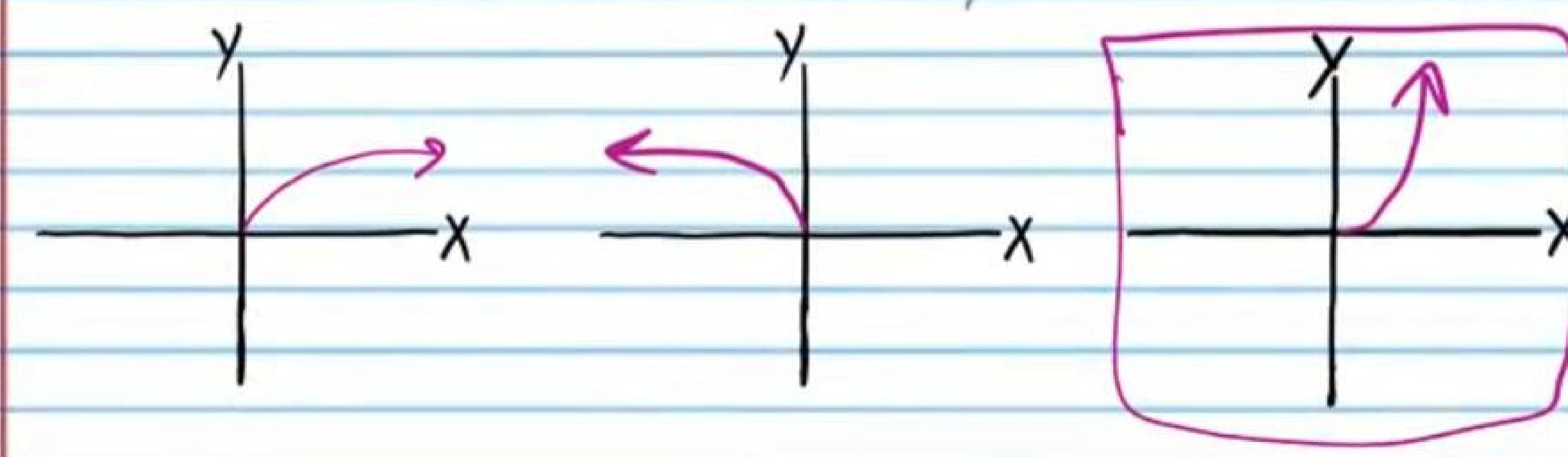


Relations are written below in the form of graphs. Draw the graphs of the inverse relations (i.e. reflect across the line $y=x$).

(101) Write the inverse relation i.e. Reflection across $y=x$ →

Step 1 — Step 2

Reflect across the y-axis — Rotate 90° clockwise

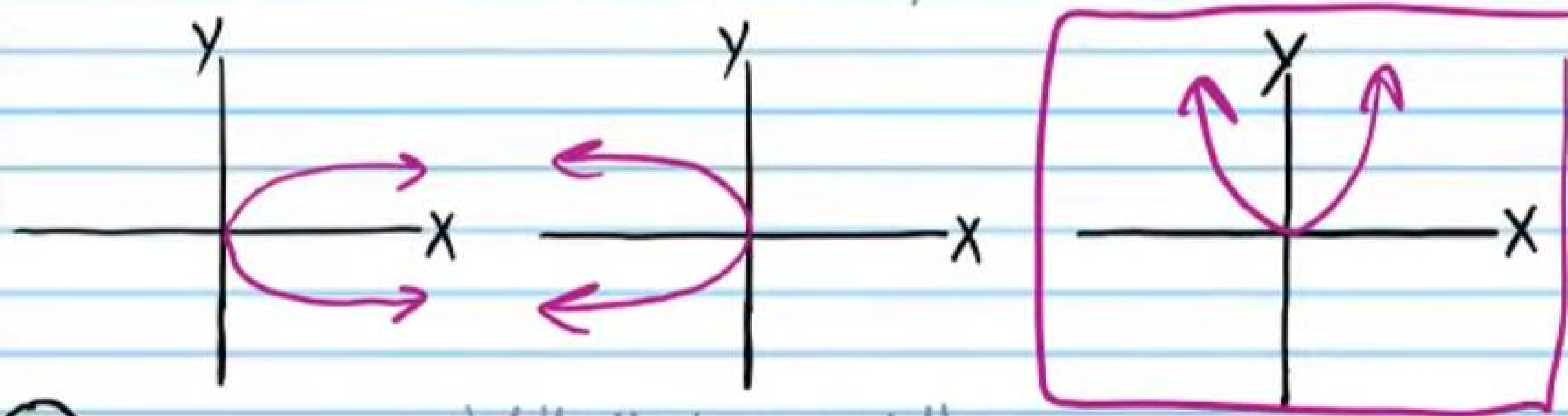


(102)

Write the inverse relation
i.e. Reflection across $y=x$

Step 1
Reflect across the y-axis

Step 2
Rotate 90° clockwise

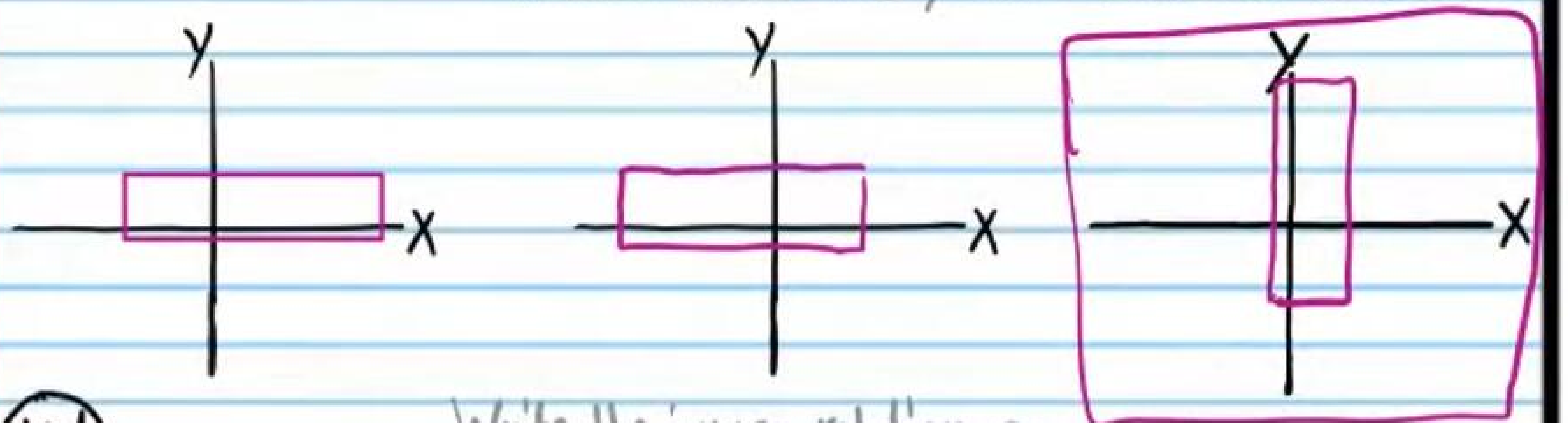


(103)

Write the inverse relation
i.e. Reflection across $y=x$

Step 1
Reflect across the y-axis

Step 2
Rotate 90° clockwise

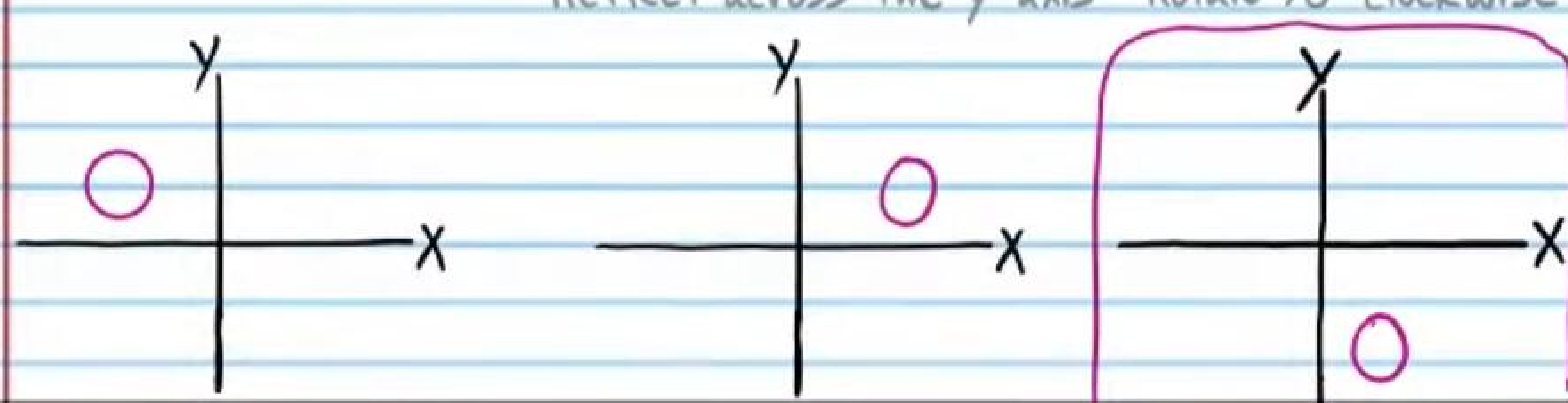


(104)

Write the inverse relation
i.e. Reflection across $y=x$

Step 1
Reflect across the y-axis

Step 2
Rotate 90° clockwise

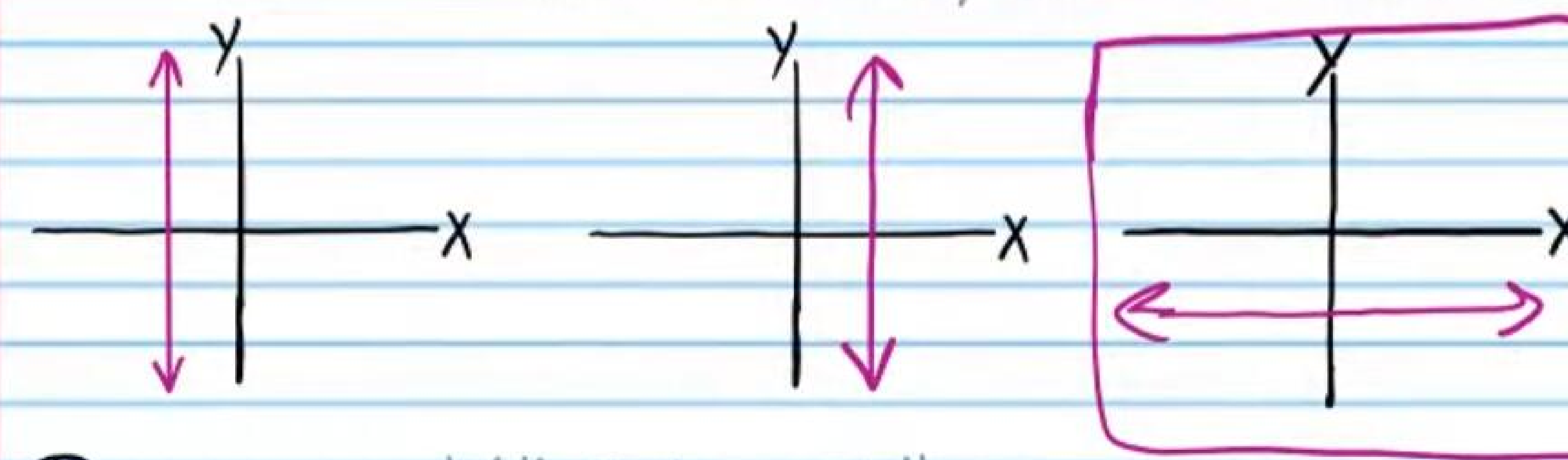


(105)

Write the inverse relation
i.e. Reflection across $y=x$

Step 1
Reflect across the y-axis

Step 2
Rotate 90° clockwise

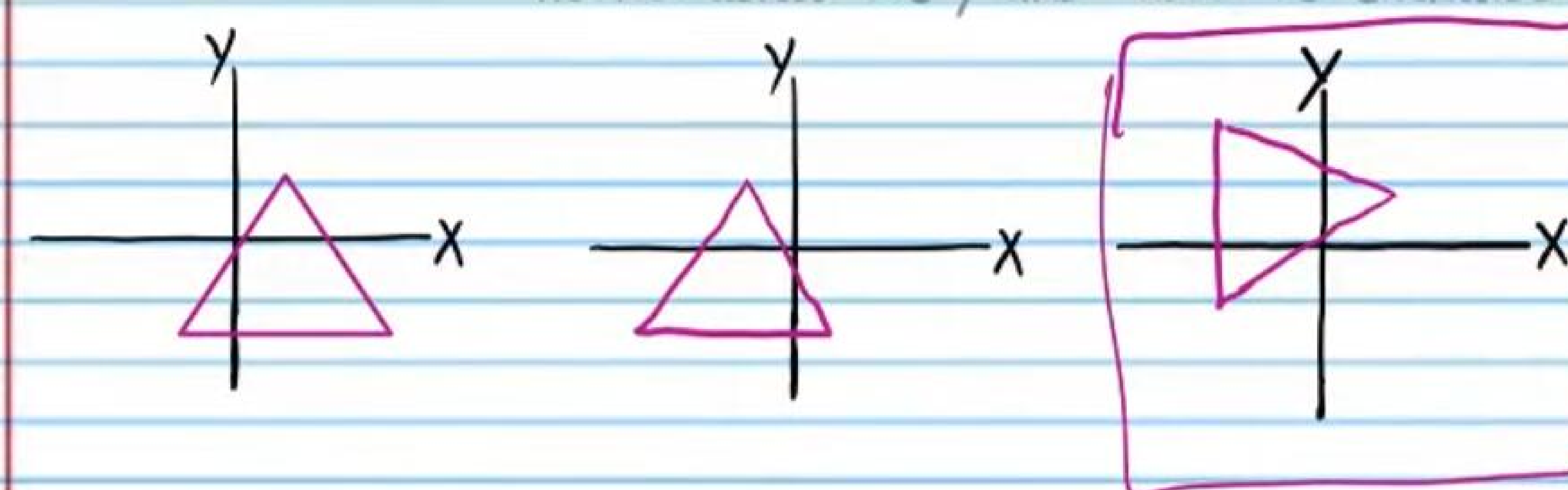


(106)

Write the inverse relation
i.e. Reflection across $y=x$

Step 1
Reflect across the y-axis

Step 2
Rotate 90° clockwise



Inverse Equations

The equation of an inverse relation can be found by simply exchanging the x and y variables of the original equation.

Example VIII

$$y = \tan x$$

Exchange the x and y variables.

$$\Downarrow$$

$$x = \tan y$$

The equations of relations are written below. Write equations of the inverse relations.

(107) $x = y - 2$ $\xrightarrow[\text{i.e. Reflection across } y=x]{\text{Write the inverse relation}}$ $\boxed{y = x - 2}$

(108) $y^3 = x^2 + 1$ $\xrightarrow[\text{i.e. Reflection across } y=x]{\text{Write the inverse relation}}$ $\boxed{x^3 = y^2 + 1}$

(109) $y^2 = \sin(x+10)$ $\xrightarrow[\text{i.e. Reflection across } y=x]{\text{Write the inverse relation}}$ $\boxed{x^2 = \sin(y+10)}$

(110) $\log_2 x = \sqrt{y}$ $\xrightarrow[\text{i.e. Reflection across } y=x]{\text{Write the inverse relation}}$ $\boxed{\log_2 y = \sqrt{x}}$

(111) $y = x^5$ $\xrightarrow[\text{i.e. Reflection across } y=x]{\text{Write the inverse relation}}$ $\boxed{x = y^5}$

(112) $3^{y-2} = x - 6$ $\xrightarrow[\text{i.e. Reflection across } y=x]{\text{Write the inverse relation}}$ $\boxed{3^{x-2} = y - 6}$

(113) $\cot y^4 = 1 + |x|$ $\xrightarrow[\text{i.e. Reflection across } y=x]{\text{Write the inverse relation}}$ $\boxed{\cot x^4 = 1 + |y|}$

(114) $\frac{1}{y^5} = \ln x$ $\xrightarrow[\text{i.e. Reflection across } y=x]{\text{Write the inverse relation}}$ $\boxed{\frac{1}{x^5} = \ln y}$